Embankment mower

Type FME 500

Translation of the Operating Manual

Edition 01/2016

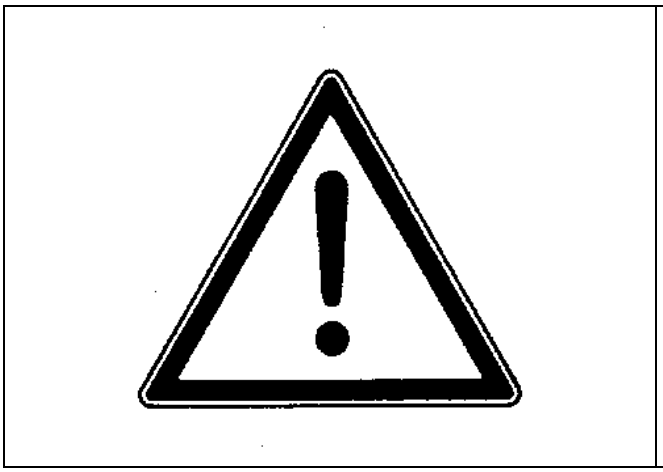


Serial number: S059972

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Operation Manual

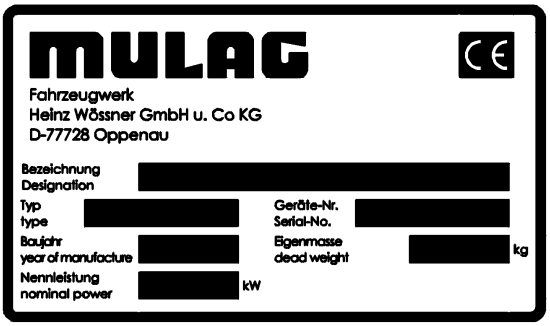
en



Always read and observe the operating instructions and safety notes prior to putting the attachment into use.

The operating manual must always be kept in an accessible location in the carrier vehicle.





Remark: The marks must be regarded as documents and may not be changed or made indecipherable

The typeplate of the Embankment Mower FME is fitted to the turned part of the boom.

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0 Preface

- This operating manual is designed to help familiarise you with the embankment mower and its intended applications. It contains important information on how to operate the mower safely, correctly and economically. Compliance with the operating manual will help you to avoid hazards, cut repair costs and downtime and increase the reliability and lifetime of the device.
- In addition to this manual and the compulsory accident prevention regulations, the recognised technical standards for safe and professional working practices must also be observed, together with the operating manuals of the relevant carrier vehicle and basic unit.
- The operating manual must be supplemented with instructions based on the existing national regulations of the user country with regard to accident prevention and environmental protection.
- The numbers in brackets after the designation of details in the text, e.g. (Figure 1/6), refer to the item number (6) on the relevant diagram (1).
- The following symbols are used in the manual in order to highlight particularly important information:



This warning symbol is used to identify all passages in the manual concerning your safety.
These instructions must be followed closely in order to avoid the risk of personal injury and damage to property.



This symbol refers to information which must be paid special attention in order to ensure trouble-free operation of the device.

0.1 Commissioning in the USA



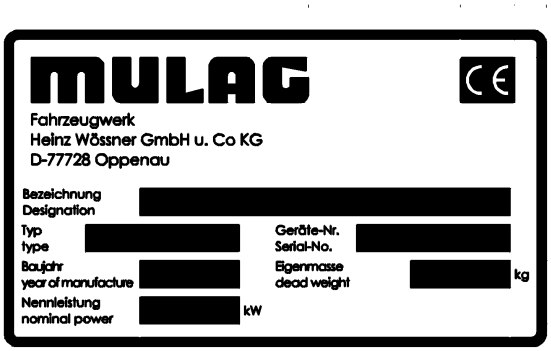
- If this device is intended for use in the USA, the operator is responsible for stipulating and adhering to the safety and precautionary measures.
- The following operating instructions serve as a guideline only and do not identify all of the known dangers and hazards.

0.2 Copyright

- All rights, including publication (complete or partial), distribution, adaptation and translation absolutely require the express written permission of the MULAG.

- Subject to technical alterations.
- This documentation was provided exclusively for internal use for training and handing over purposes by MULAG customers.
- Address:
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0.3 Device-identification



- All labels and tags must be regarded as documents and may not be changed or made illegible.
- The type plate of the Type MFK is located on the pivot pin of the main boom.

Following information are stamped on the type plate:

- Name of the machine
- Machine type
- Year of manufacture of the machine
- Output in kW
- Device identification or serial number
- Weight of the equipment

0.4 Decoding the machine number

- The equipment identification or serial number consists of following components:

  16 / S / 039535

- The first two figures indicate the year of manufacture of the machine.
- The letter S indicates that this equipment belongs to the SUG activity (road maintenance) of the MULAG production range.
- The last six numbers are a computer-generated serial number or MULAG batch number for the identification of the equipment.
- For the identification of all components, which were not manufactured by MULAG, identification plates are also attached on the component parts, giving an identification and an allocation on year of manufacture, month and serial number of the components.

0.5 Copy of the declaration of conformity

MULAG

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D-77728 Oppenau

EC Declaration of Conformity

We hereby declare that, due to its build and type of construction, as well as the design in which we market it, the machine designated below conforms to the relevant basic health and safety requirements stipulated by this EC Machine Directive. Any modifications of this machine which are not first agreed with us will void this declaration.

Instructed with the compilation of technical documents:	Mario Matz Heinz Wössner GmbH u. Co. KG Gewerbestraße 8 D-77728 Oppenau
Make:	**MULAG**
Machine Designation:	MFK 500, MFK 500 T, MFK 400, FME 500, FME 600, FME 600 T, ME 700, SB 500, SB 600, MHU 800, MHS 800, MRM 300, MRS 300, MRK 300 G
Machine no.:	_____
Relevant EC Directive:	EC Machine Directive (2006/42/EG) EC Directive of electromagnetic compatibility (2014/30/EU)
Applicable harmonised standards:	DIN EN ISO 12100, DIN EN ISO 13857, DIN EN 349, DIN EN 13524, DIN EN ISO 13849, DIN EN ISO 4413
Applicable national standards and technical specifications:	DGUV Regulation 113-015

Oppenau, 20th May 2016

Holger Wössner
Managing Director

1 Safety

1.1 Intended purpose

- The embankment mower is intended exclusively for industrial use, in order to carry out forestry and municipal clearing work on consolidated roads.
- The embankment mower is designed to be attached to municipal or agricultural carrier vehicles.
- While the embankment mower is being used for its intended purpose (e.g. mowing grass verges and embankments), the carrier vehicle may only travel on properly consolidated paths and roads.
- The above work may only be performed using devices which have been either manufactured or approved by MULAG.
- Any other usage constitutes a breach of the intended purpose, and MULAG accepts no liability for damage resulting therefrom; the risk in this case is borne entirely by the user.
- Compliance with the intended purpose also includes observance of the operating conditions prescribed by MULAG and adherence to the servicing and maintenance regulations.

1.2 Operating safety

- The embankment mower is built to state-of-the-art standards and is safe to operate. However, it cannot be fully protected by structural features without impairing its function and operability. As a result, the user or a third party may be exposed to risks to life and limb if the device is not used by trained personnel and/or used improperly or for purposes other than that for which it was designed.

- All persons employed in the assembly, start-up, operation, maintenance and repair of the embankment mower must have read and understood the operating manual and in particular this "Safety" chapter. The employed personnel must confirm in writing in the delivery declaration that they have participated in an induction or training course provided by an authorised representative of MULAG.

- All notices affixed to the embankment mower and basic unit must be observed and the instructions printed on them followed.

- The driver of the carrier vehicle must be aware of the changes brought about by the embankment mower in the driving quality of the carrier vehicle before using it on public highways.

- The responsibilities for the various jobs associated with the embankment mower must be clearly defined and observed in order to rule out any uncertainty with regard to safety. This applies in particular to work on the electrical and hydraulic equipment, which may only be performed by suitably qualified specialists.

- The operating manual for the embankment mower must be kept handy on the carrier vehicle.

- Access to the danger zone around the boom is prohibited during operation.

- Safety devices must not be dismounted or taken out of operation. Non-compliance may lead to risks to life and limb. If safety devices have to be dismounted for set-up, repair or maintenance purposes, they must be re-mounted immediately on completion of the work.

- The safety devices provided must be mounted for transportation or return journeys.
- If the safety devices become worn or lost, work must be suspended immediately and the safety devices replaced.
- The customer must ensure that the device is only operated when in sound working condition.
- The embankment mower may only be operated by trained and authorised personnel.
- All working practices detrimental to the safe handling of the embankment mower are prohibited.
- The operator must ensure that the embankment mower is only used by authorised personnel (e.g. by locking to prevent unauthorised use).
- The operator is obliged to check the embankment mower at least once a day for externally visible damage and defects. Any changes (also in its operation) which may pose a safety hazard must be rectified immediately.
- A defective embankment mower must not be used or put into operation.
- The device may only be used in the direction of travel, i.e. the boom must be able to move clear to the rear.
- Unauthorised conversions or modifications to the embankment mower are prohibited for safety reasons.
- While the carrier vehicle is approaching the embankment mower during the set-up procedure, keep clear of the danger zone around the vehicle and mower in case the mower is knocked off its storage supports by the vehicle.
- Take note of and allow for the changed dimensions of the carrier vehicle due to the mower device.

- The device and/or boom must be removed and secured for long-distance and transportation journeys.
- During long-distance and transportation journeys, the operating levers of the hydraulic system on the carrier vehicle and embankment mower must not be actuated.
- Only start up the embankment mower when there is no-one present in the danger zone. While the device is in operation, there is a risk of injury from ejected objects such as stones, bits of wood, grass and undergrowth remnants. The entire slewing range of the boom also constitutes a danger zone.
- During operation, the cab door and windows on the working side must be closed. Leaving the cab door and/or window open exposes the driver to a risk of injury.
- Do not turn the carrier vehicle round when the boom is extended in case it tips over.
- When working on gradients, make sure the stability of the carrier vehicle is maintained. The camber of the road may not exceed 5%. On stretches with both a downhill gradient and a camber, it may be necessary to reduce the working range.
- The working speed may not exceed 10 km/h.
- If the embankment mower hits a power cable, the driver/passenger must remain in the closed cab until the device is no longer touching the cable or until the cable has been de-energised.
- Whenever the device is idle, e.g. during breaks or after finishing work, the electrical control system must be switched off via the operating switch/key switch and the power transmission/hydraulic supply to the embankment mower must be shut down via the PTO shaft control lever.
- During lengthy idle periods, the key switch must be removed and kept in a safe place.

- When equipping the carrier vehicle with an embankment mower and the associated attachments, pay attention to the permissible axle loads and overall vehicle weight.
- If the vehicle number plate or lighting is obscured by the attachments, it must be re-affixed according to the regulations (e.g. highway code).
- It must be ensured that the combination of carrier vehicle, basic unit and mower head corresponds to the applicable regulations of the country where it is operated (e.g. highway code, permissible total weight, etc.).
- If optional extras or attachments are subsequently fitted, make sure they do not affect the stability of the carrier vehicle.
- When approaching crossroads/junctions, be aware of the changed dimensions of the vehicle due to the attachments.
- Only set down the embankment mower in a stable position on the storage supports provided. Always make sure it is deposited on a horizontal, properly consolidated surface.
- Operating MULAG mowers requires big concentration. That is the reason why - when organizing breaks to remain concentrated - MULAG recommends to observe at least the legal steering and rest periods for vehicles over 7,5 t. This is valid also for vehicles under 7,5 t total permissible weight.
- **Definition danger zone:**
  The danger zone around the mowing equipment is always: "**7 m in the swivelling area of the arm**", but it can be increased considerably according to the working implement used (flinging stones etc). Do observe the valid safety zone described in the operating manual of each working attachment.
- Attention must be paid that no combustible objects and no combustible liquids or operating fluids can get onto heated units such as motor, transmission, exhaust system, turbocharger or similar.

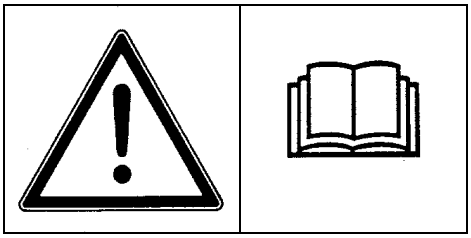
- To clearly reduce the risk of ignition by hot vehicle components or parts of the structure, the carrier vehicle together with the base unit and working implement must be cleaned and maintained on a regular basis. In particular grass, plant matter and dust deposited in the chassis and structure must be removed regularly.
- Other road users must be warned by safeguarding the danger zone in accordance with the corresponding traffic law regulations currently in force.

1.3 Safety during servicing and maintenance

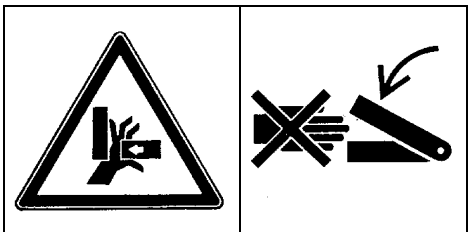
- Servicing, maintenance and inspection work must be performed at the recommended intervals (sections 7.2 & 7.3).
- Only allow suitably authorised specialists to carry out servicing, maintenance and inspection work, particularly on the electrical and hydraulic equipment.
- Always switch off the engine of the vehicle before performing any work on the embankment mower.
- Damaged or worn parts of the embankment mower must be replaced immediately. For reasons of safety and product liability, only original MULAG spare parts may be used, otherwise the safety of the device according to the machinery directive 2006/42/ EG is no longer guaranteed.
  All improvements – or even attempted improvements – to the product or combinations of the product with spare parts not supplied or explicitly approved in writing by MULAG which are performed by third parties or the customer without our written consent will lead to the loss of all guarantee claims. This does not apply if the customer can prove that a defect in the product is not due to the above actions.
- When changing wearing parts, the corresponding fixing devices, e.g. screws, nuts, etc., must be replaced with new or equivalent ones.
- Before carrying out any work on the hydraulic system, the cylinders and pipes must be relieved of pressure in order to prevent the risk of injury, explosion and fire. The system is subject to high pressures and temperatures. Even a very fine oil jet can penetrate the skin and cause serious injuries. Should this happen, a doctor should be consulted immediately.
- Damaged or missing notices and warning signs must be replaced immediately.
- Maintenance and repair work may not be performed on public highways, due to the increased risk of accident and contamination.

- After care/maintenance and inspection work, make sure all dismounted protective devices and locking elements are correctly reassembled.
- Hydraulic hose-pipes must be monitored according to the national safety regulations for hydraulic hose-pipes.
- When washing the embankment mower, do not aim the water jet at electrical or hydraulic devices (e.g. filters, couplings, tank filler neck) or conductors.
- When selecting lubricants, hydraulic oils and filter cartridges, pay attention to environmental compatibility, health risks, disposal regulations and your available local facilities for safe disposal. The applicable laws and regulations and the manufacturer's guidelines must be observed.
- If oil escapes onto the road surface, stop work immediately, seal the leak and absorb the escaped oil with binding agent. Take the necessary safety measures and have the defect repaired.
- Do not bring hydraulic oil into contact with mucous membranes, eyes, nose or mouth. Should this happen, consult a doctor immediately.
- Only carry out care/maintenance and inspection work on properly consolidated ground intended for the purpose (with oil separators in the drainage system).
- In the event of modifications or additions to the electrical equipment of the machine, the owner (or company performing the modification) must make sure the electromagnetic compatibility of the overall system is maintained (EMC directive).

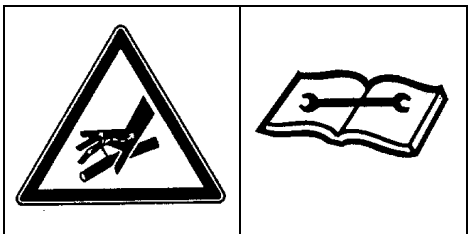
1.4 Notices



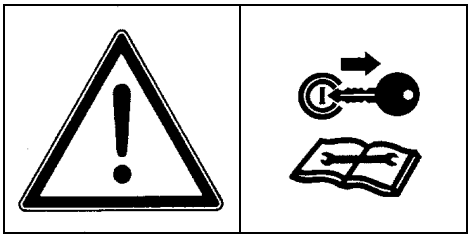
- Read and observe the operating manual and safety information before start-up.



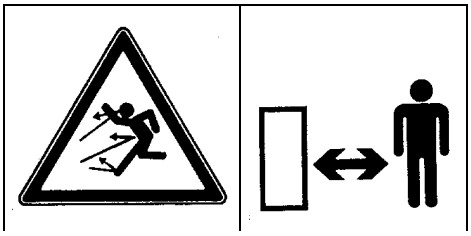
- Never reach into the crushing zone while parts may be moving inside it.



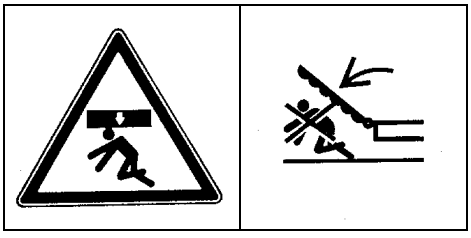
- Caution when handling high-pressure fluid leakages. See note in technical manual.



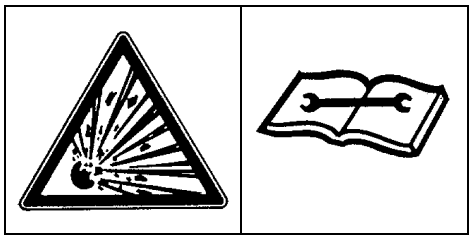
- Switch off the engine and remove the key before maintenance and repair work.



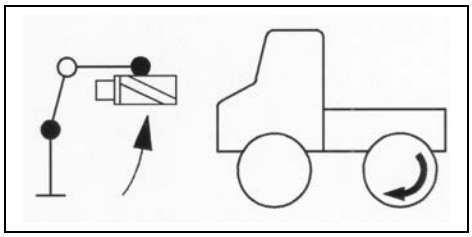
- Danger due to ejected matter when engine is running – keep a safe distance.



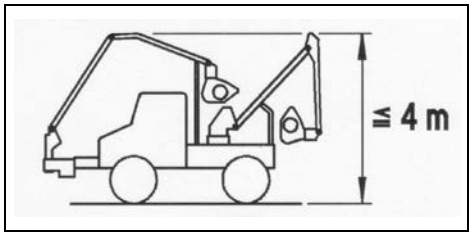
- Keep out of the slewing range of the equipment.



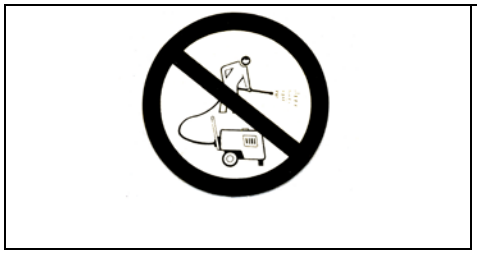
- Accumulator under gas and oil pressure. Only dismount and repair according to instructions in technical manual.



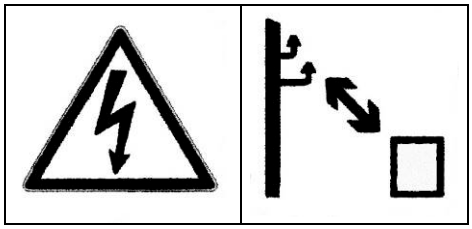
- Never reverse the vehicle when the mower device is switched on.



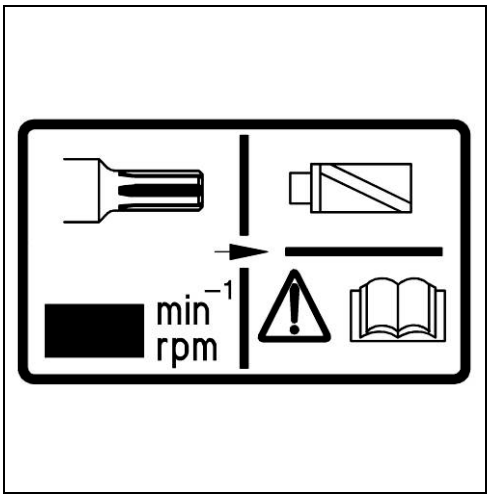
- Always observe the correct transportation height when the mower device is in the parked position.



- By cleaning works with a high-pressure equipment, be careful not to orientate the steam jet directly on the switchbox or control block.



A sufficient safe distance to electrical high voltage lines must be kept by.
Minimum distance 5 m!



- Set the required power take-off shaft speed with the corresponding motor speed so that the cutter shaft speed is reached.
- The cutter shaft speed is specified in the chapter "Technical specifications" in the operating manual of the working implement.

2 Technical data



- Observe technical data. Maintain pressure and flow rate of hydraulic oil.
- The required volumetric flow depends on the type of working attachment (please consult the operating manual of the working implement).

Upward and downward slewing range of boom (respectively): ..... 45°

With offset boom attachment for upward direction: ..... 110°

to right and left FME 400/500 (respectively): ..... 106°

Max.range from centre of vehicle to right / left
FME 500 : .....6.5 m

Weight: ..... see nameplate

Working speed depending on type of grass
and composition of verge: ..... 1-10 km/h

Hydraulic tank capacity: .....approx.60 l

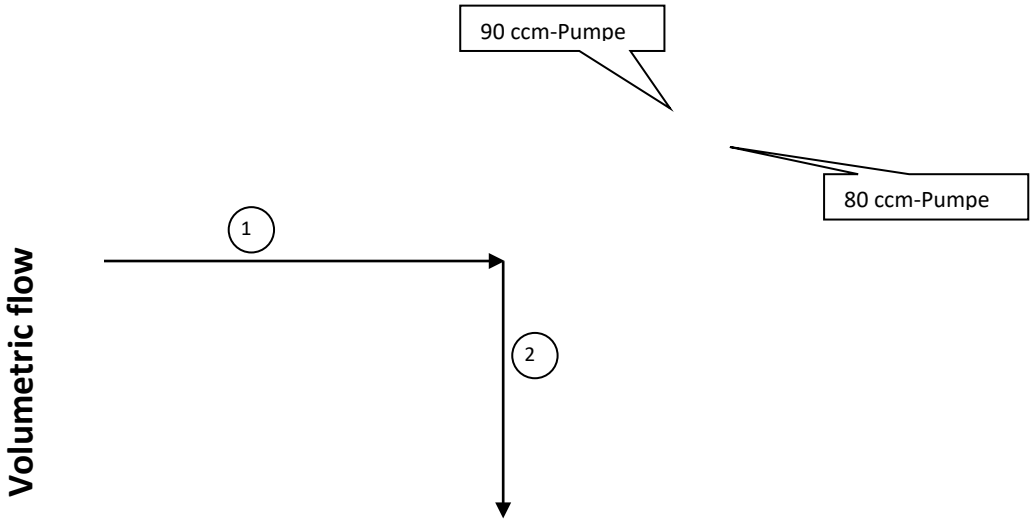
Hydraulic oil pressure, mower drive: .....$340^{\pm 5}$ bar

Hydraulic oil pressure and pressure limiters: .....see section 9.1

Required hydraulic equipment (carrier vehicle supply):
Hydr.oil pressure: ..... 200^{+20} bar

Volumetric flow of hydraulics: .....35-40 l

2.1 Volumetric flow diagram



Carrier vehicle	PTO gearbox Selector [rpm]
U 130 – U 2450	1000
JCB Fastrac 135 – 185	1000
JCB Fastrac 2115 – 2135	1000
JCB Fastrac 3155 – 3185	1000
Fendt 380 GTA / 395 GTA	1000
Fendt Xylon 520 – 524	1000

2.2 Equipment driven via power hydraulic system

- When operating the equipment via hydraulics of the vehicle control the cutter shaft speed of the mowing heads thanks to a mechanical manual speedometer. For this purpose see the operating manual of the working implement.

☞

- Depending on the carrier vehicle some important deviations may appear between the volume flow indicated and the volume flow which is effectively available. A too high rotation speed can damage the mowing head.
- Changing the drive via power hydraulics from closed circuit over to the open circuit and vice-versa may also cause damage and IS FORBIDDEN if our MULAG service department has not been consulted.

- Should the equipment be driven via a power hydraulic system please observe following technical data of the carrier vehicle:

• Hydraulic oil pressure when connecting the command :	Min. 180 bar Max. 200 bar
• Flow rate when connecting the command:	40 l/ min.
• Hydraulic oil pressure when connecting the working device	min. 340 bar
• Flow rate when connecting the working device :	Depending on the working device

2.3 Emission values (sound pressure level)



The emission values (sound pressure level dB(A)) are below 70 dB(A). The noise pollution depends on the type of basic unit and carrier vehicle.

3 Description

3.1 Description of equipment

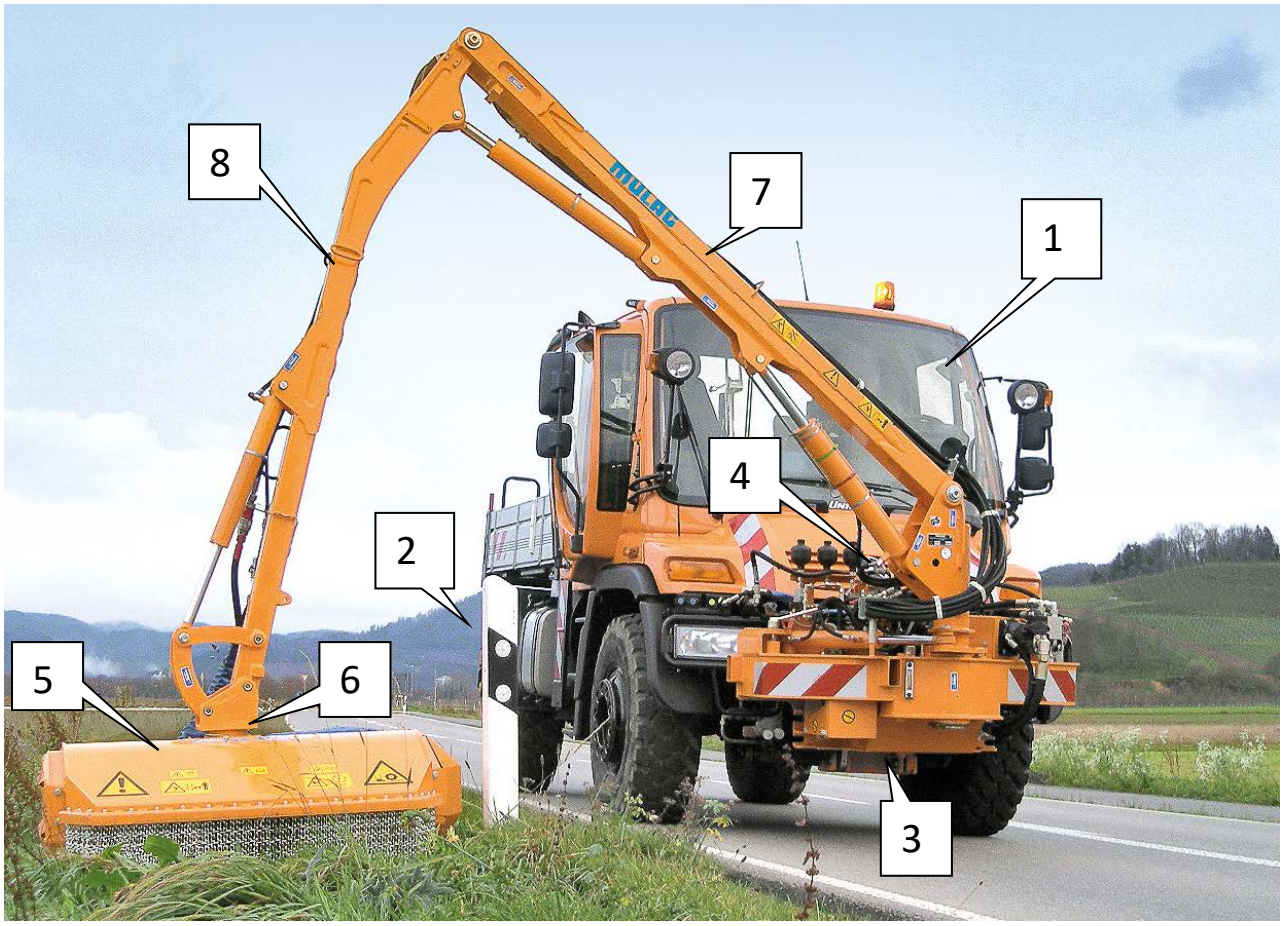


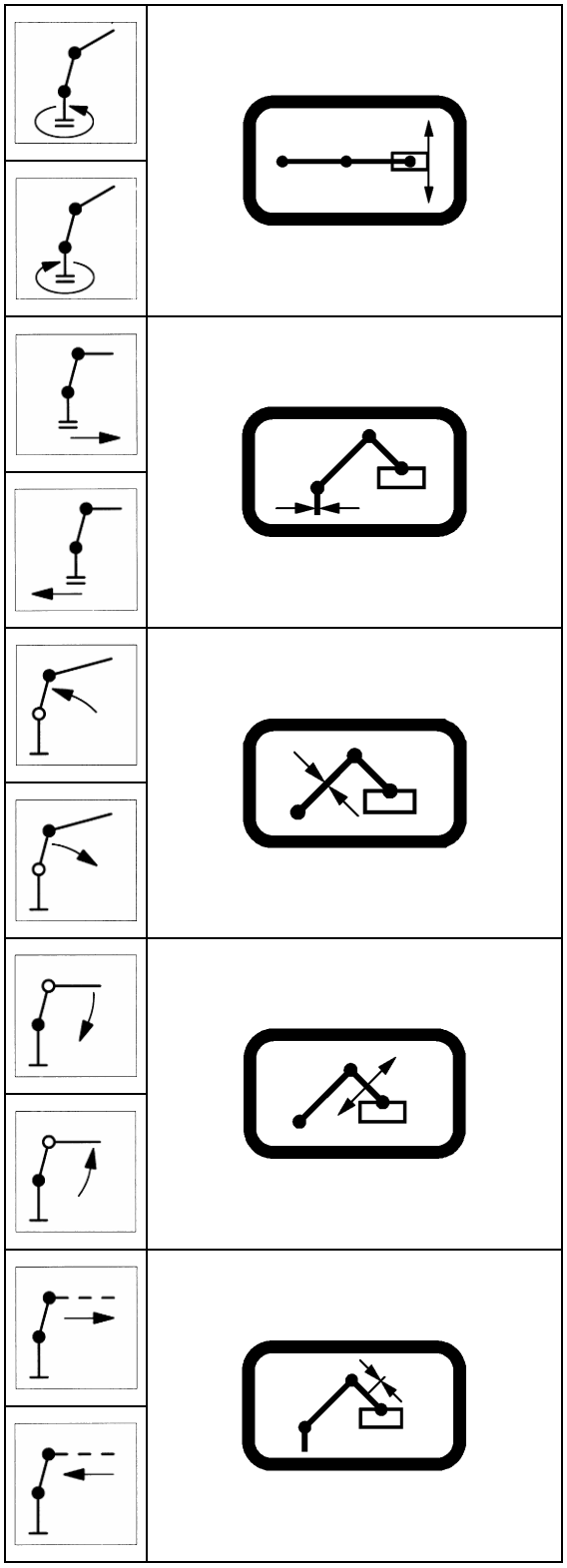
Figure 1

1. Carrier vehicle
2. Ballast weight (without figure)
3. Torsion frame
4. control block
5. Mower device
6. Tool quick-change device
7. Boom 1
8. Boom 2

- The embankment mower is designed to be mounted on the front of a carrier vehicle (Figure 1/1). A hydraulically controlled boom (Figure 1/7, 8) with three articulated joints and one pivot guides the mower device to its working position.
- While the carrier vehicle is travelling along consolidated paths or roads, the mower device (Figure 1/2) can mow both flat surfaces and slopes/embankments and the vegetation growing on them up to its maximum range.
- The ballast weight (Figure 1/2) serves to compensate for the forces generated by the boom and the torsion frame (Figure 1/4) increases the vertical stability of the vehicle.
- The parking frame (Figure 1/2) (option) is for storing and securing the boom during transportation journeys.
- Uneven ground is compensated for by a blade head floating position control system “MKS” at the sides and by a boom balancing control system “AES“ or “AER (option) in the direction of travel.
- The fully automatic non range-dependent boom balancing control system “m|tronic“ (option) automatically regulates the contact pressure and tilting angle of the mower device.
- The cuttings remain in situ.
- The embankment mower is suitable for right and left-hand mowing.
- For journeys to and from the site of operation, the boom or mower device is placed on the parking frame, transportation support or loading surface of the carrier vehicle.
- In order to perform right-hand mowing on a carrier vehicle with a left-hand drive, it is advisable to use two people (one to drive the vehicle and one to operate the mower device).
- For left-hand mowing, the embankment mower can be operated by the driver himself.

- Thanks to the quick-change device, it is possible to exchange the flail-type mower head for various MULAG devices.
- If the carrier vehicle is to be used for other purposes, the entire mower device can be removed and deposited on three supports.
- The mower device is operated via control devices mounted in the carrier vehicle.
- The operating elements are clearly assigned to the functions by means of symbols.

3.2 Graphic symbols



Swivel boom to left

Swivel boom to right

Move boom sideways to the right

Move boom sideways to the left

Raise boom arm

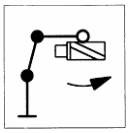
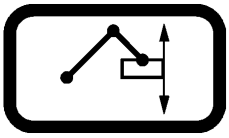
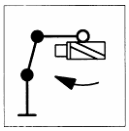
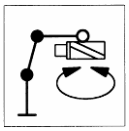
Lower boom arm

Bend boom arm

Straighten boom arm

Extend telescope

Retract telescope

		Tilt the working attachment outwards
		Tilt the working attachment inwards
		Rotate implement

4 Assembly of the embankment mower

4.1 Mounting the embankment mower



- In order to avoid the risk of overturning and damaging the vehicle, some carrier vehicles require the use of a MULAG torsion frame (option) to reduce the one-sided spring deflection of the front axle.
- While the carrier vehicle is approaching the resting embankment mower, it is important to keep a safe distance of 7 m from the device in case it is accidentally knocked over.
- In the case of vehicles with a tilting platform, the function “Tilt platform“ must be deactivated as described in the carrier vehicle manual.
- Do not actuate the control unit of the device until the hydraulic hose-pipes have been connected to the hydraulic system.
- Make sure connections are secure, otherwise the control unit may be damaged.



Figure 2

- Drive the carrier vehicle carefully towards the embankment mower. This task should only be performed using a second person as a guide (observe safety distance!) due to the risk of overturning.

- Always switch off the motor and hydraulic system before carrying out any work on the embankment mower.
- Secure the carrier vehicle to prevent it from starting up or rolling away accidentally.
- Remove the key from the key switch in the operating panel.

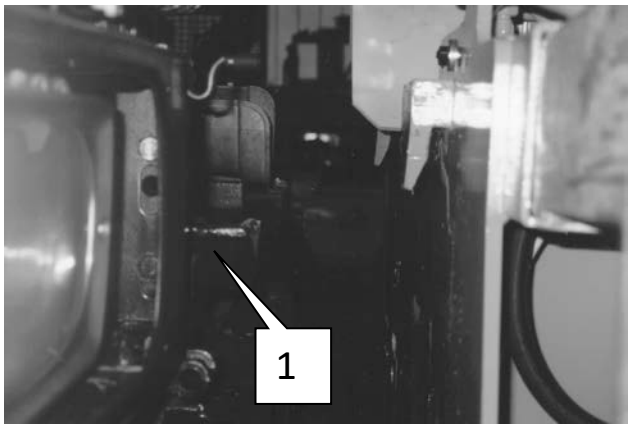


Figure 3

- Centre the vehicle over the claw pockets of the front mounting plate (Figure 3/1) until the claws of the device are directly above the claw pockets.
- Switch off the engine.

4.2 Assembling the hydraulic system

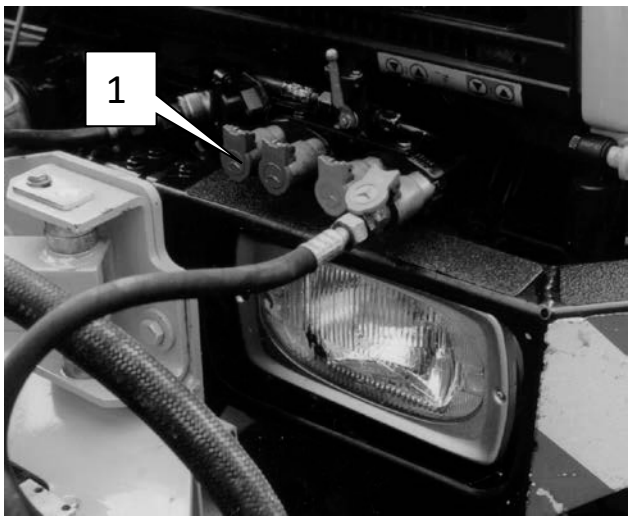


Figure 4



Figure 5

- Remove the protective caps of the hydraulic couplings (Figure 4/1) and connect the hydraulic pipes. For cell arrangement and symbols, see operating manual and technical data of carrier vehicle.
- Connect the non-pressurised return hose first, then the delivery hose.

- Lower the three supports of the embankment mower evenly by turning the crank handle until the claws engage in the claw pockets of the front mounting panel.

When lowering the embankment mower, actuate the control lever for the function **“Lift/lower boom“** every now and again. It is essential to keep a safe distance while doing so, due to the danger of crushing in the region of the boom. This relieves the strain on the lifting cylinder and allows the device to be lowered more evenly. For this purpose, place the emergency hand lever from the vehicle tool set on the **“Lift/lower“** segment. Once the device is mounted, the lever must be removed and stored in the vehicle tool set in order to prevent the device from being operated by unauthorised persons.

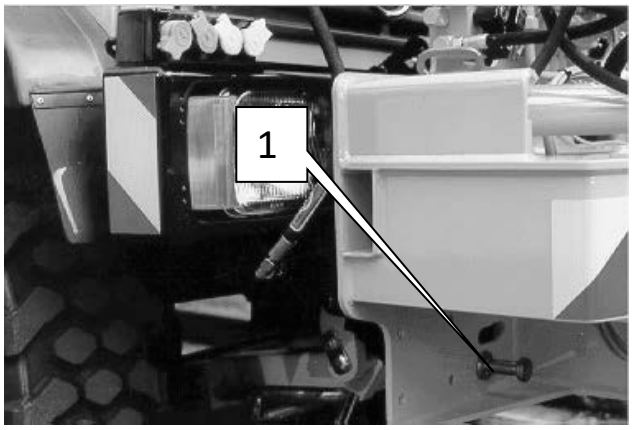


Figure 6

- Fasten the front frame at the bottom with two collar screws (Figure 6/1) or lock with a “Servolock“ device as described in the operating manual of the carrier vehicle.
- Remove the supports from the embankment mower and keep safe.

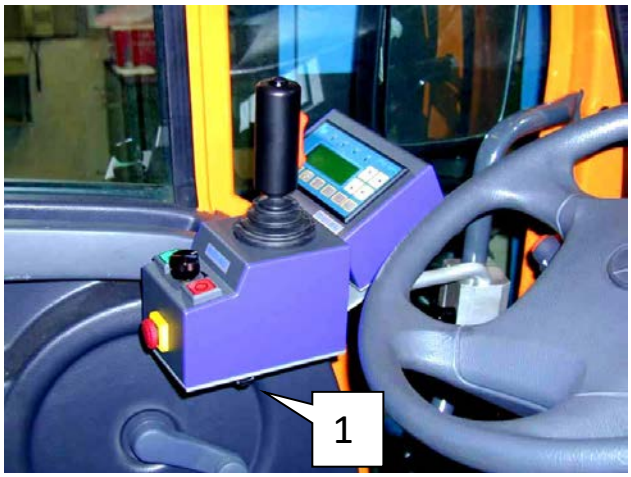


Figure 7

- Lay the electrical cable to the operating panel (Figure 7/1) so that there is no risk of tripping over it.
- Plug the electrical connection in the driver’s cab into the cigarette lighter or the service current socket (option), depending on the relevant model.

4.2.1 Assembling the hydraulic pump unit FME 400/500

- Make sure the locking pin (Figure 8/1) of the cardan shaft or double joint clicks into place when connected.
- After completing the following adjustments, mount the protective cover of the PTO shaft with the opening facing downwards so that the PTO shaft connection is covered.

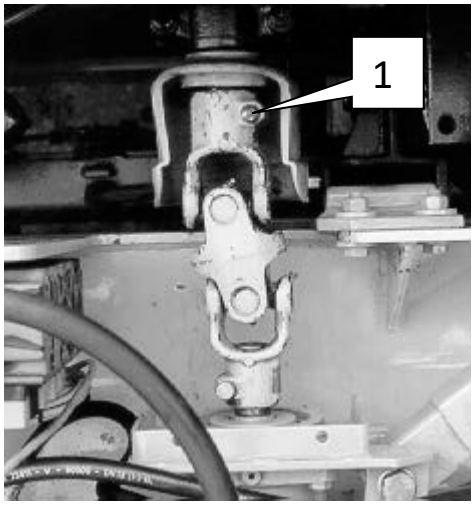
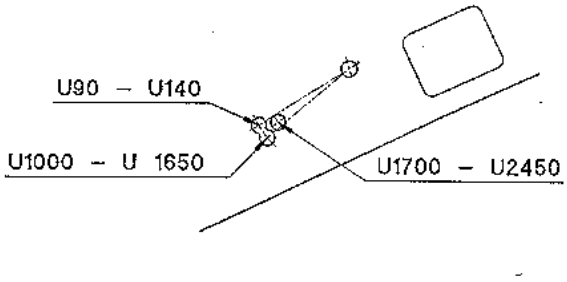


Figure 8

- Connect the cardan shaft (Figure 8/1) of the embankment mower to the PTO shaft of the carrier vehicle.

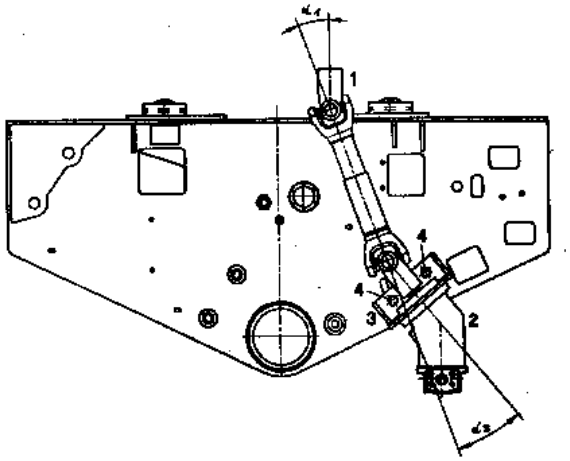
The setting of the cardan shaft (joint angle) depends on the type of carrier vehicle. An incorrect setting will lead to premature wear and failure of the cardan shaft. The joint angles a1and a2 must be approximately equal.



- Adjust the hydraulic pump to the relevant vehicle type as follows: Insert the fastening screw for the hydraulic pump holder through the hole according to the vehicle type and tighten.

Figure 9

Unimog vehicle type



- In other types of vehicles, adjust the cardan shaft so that angles a 1 and a 2 are approximately the same. If the hole in the pump holder does not match the existing hole, a new hole must be drilled in the front frame.

Figure 10

4.3 Mounting the operating panel

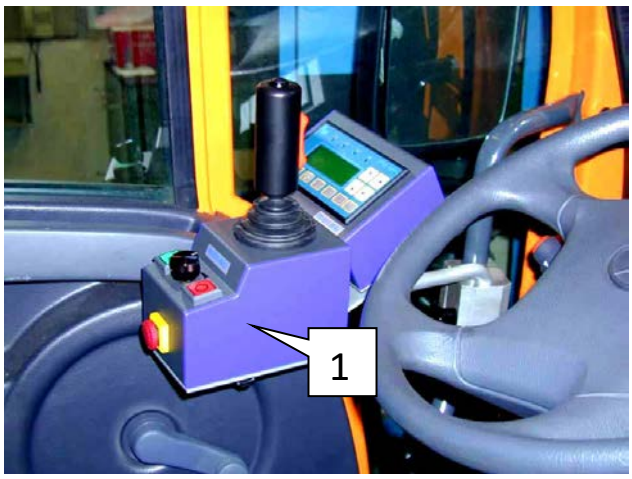


Figure 11

- Mount the operating panel in a suitable place in the driver's cab of the carrier vehicle (Figure 11/1).
- Check to make sure the electrical voltage of the carrier vehicle coincides with the rated voltage (12 V or 24 V). The voltage rating is indicated on the key switch in the operating panel (e.g. Figure 11/2).

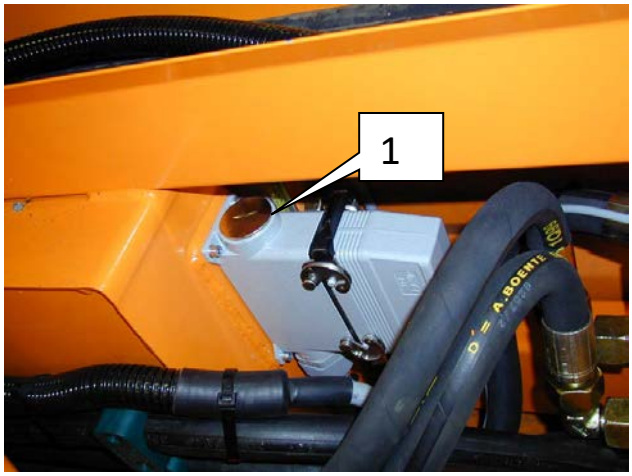


Figure 12

- Insert the control cable with the appliance connector at the connecting points and lock in place (Figure 12/1)
- Lay the electrical cable in such a way as to eliminate all risks during operation and transportation (e.g. getting in and out of the vehicle).
- Connect the operating panel to the electrical system of the vehicle via the cigarette lighter or service current socket depending on the chosen options.

4.4 Mounting the parking frame and ballast weight

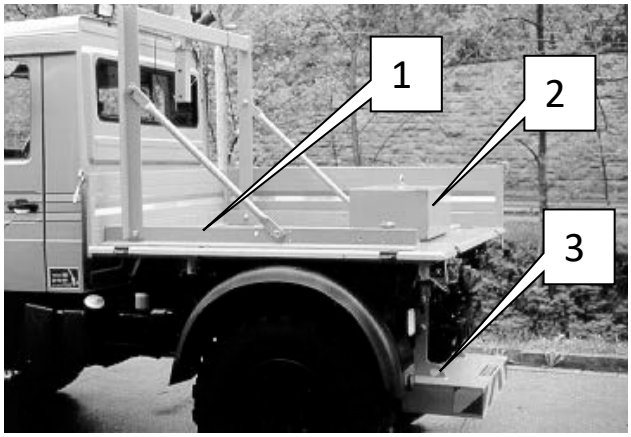


Figure 13

- Place the parking frame (Figure 13/1) at the centre of the loading surface of the carrier vehicle so that it lies in the longitudinal direction, flush with the front panel of the vehicle.
- Screw the parking frame to the loading surface using the four holes provided. Drill through the platform for purposes of initial assembly.

- Insert ballast weight (Figure 13/2) in the parking frame and screw in place at the two supporting points provided.

or

- Fix ballast weight (Figure 13/3) in the mounted fittings (D50) using four screws.

or

- Insert ballast weight in the three-point rod assembly and secure.

4.5 Ballasting through ballast weight system

- The ballast weight depends on the type of carrier vehicle.
- If the required ballast weight is not included in the standard unit, contact MULAG to know what kind of weight you need (give the specifications of the vehicle). In case of Unimog Type 427 / 437 as well as U 300 and U 400, the ballast weight will be mounted on the rear fixtures (D50) of the vehicle.

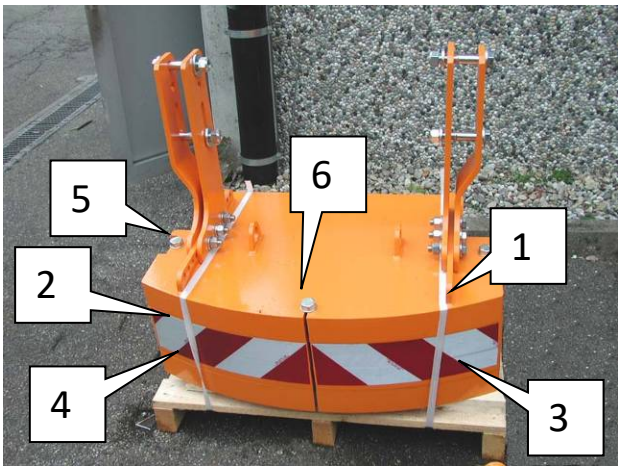


Figure 14

- For ballasting the carrier vehicle, MULAG offers a standardized ballast weight system for the Unimog models 427, 437 and U300 and U 400.

Number of weights	Item number	Total mass
1	170132	350 kg
2	170140	700 kg
3	181669	1050 kg
4	181650	1400 kg

- Refer to above charts to know the available weights and the number of ballast plates necessary.
- According to this system the weights 350 kg, 700 kg, 1050 kg and 1400 kg can be mounted with different basic weights of 350 kg respectively.

- When mounting the ballast weight or its individual weight plates, suitable lifting equipment such as a crane or a lift truck and wooden timbers must be used as support. Otherwise there is risk of accident from falling weights when the long bolts are released.
- The holding bolts for the weight plates (Figure 14/5/6) must be tightened to a torque of 639 Nm.

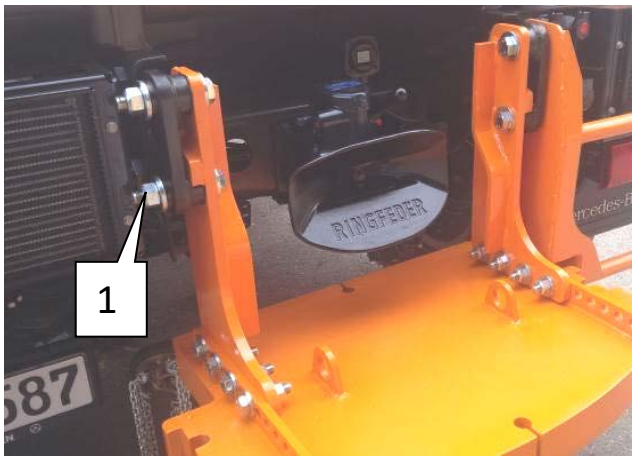


Figure 15

- Secure the ballast weight (Figure 15/1) with four bolts in the attachment fittings (D 50).

5 Start-up – application - operation

- Initial start-up should only take place after practical instruction by someone authorised by MULAG. Unauthorised attempts to start up the device can lead to personal injury and damage to property.
- Check the road safety of the carrier vehicle before driving it.
- Make sure the number plate and lights of the carrier vehicle are clearly visible and not obscured by mowing devices. If necessary, the vehicle should be equipped with an additional number plate and/or lights.
- Before long-distance or transportation journeys, the spring deflection lock on the carrier vehicle must be deactivated.



- In the case of vehicles with a tilting platform, the function “Tilt platform“ must be deactivated according to the instructions of the carrier vehicle manual.
- Before swivelling the boom from the transport to the working position, reduce the spring deflection on one side of the front axle.
  (e.g. by locking the MULAG torsion frame.)
- This spring deflection must remain locked in the reduced position until the boom is placed back on the support system and secured.
- When starting up or moving the boom to the transport position, it is advisable to reduce the engine speed, as this allows a more sensitive control of the boom system.

5.1 Putting the embankment mower into operation

Drive to an area with consolidated paths/roads where you can familiarise yourself with the functions and operating elements of your new embankment mower at your leisure.



- Make sure there is no-one in the danger zone before beginning work.
- The cab doors and windows must be closed during operation in order to protect the operator and driver against noise and any matter thrown up by the mower device.
- Always switch off the engine and hydraulic system before carrying out any work on the embankment mower.

- Lock torsion frame. Operation is described in the separate operating instructions.

5.2 Switching on the hydraulic system

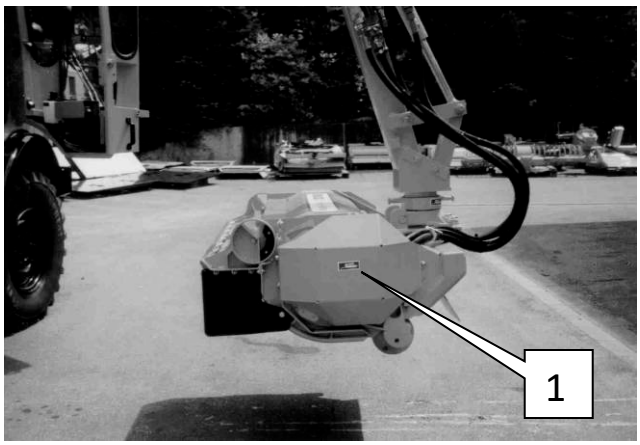
- Switch on the hydraulic cell connected to the mower according to the operating manual of the carrier vehicle.
- Plug the power supply cable of the control unit into the service current socket or cigarette lighter of the carrier vehicle.
- Start up the engine of the carrier vehicle. Select a reduced engine speed.

5.3 Swivelling the mower device to the working position

- Release boom from parking frame or remove second swivel limit stop.
- Open ball cock for **"Swivel mower device"** function. If the mower device rests on transport supports, also open the ball cocks for **" Lift boom"** and **" Bend boom"**.

> ☞ The embankment mower cannot perform any functions when the ball cocks are closed.

- Start up the engine and preselect a low speed.
- Switch on the hydraulic system.
- Switch on the voltage supply via the switch in the operating panel. (All functions must be deactivated, otherwise the operating panel cannot be switched on.)
- Move boom 1 to the vertical position via the **"Lift"** function.
- Swivel the boom system to the working position via the **"Swivel/extend boom"** function.
- Remove transport support (if applicable).



- Move the mower device (Figure 16/1) to a horizontal position close to the vehicle with a ground clearance of approx. 10 cm.

Figure 16

- As soon as the ring (Figure 17/2) is pushed up, the mower device can rotate of its own accord, thus posing a risk of accidents.
- Observe safety instructions in mower device manuals.

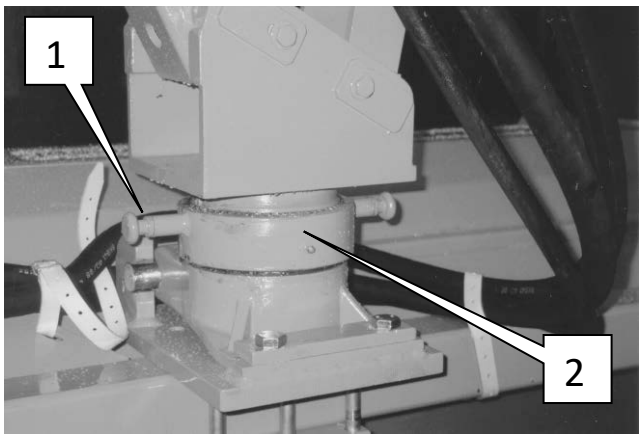


Figure 17

- Pull both locking pins (Figure 17/1) simultaneously and push up ring (Figure 17/2).
- Rotate mower device to working position.
- Pull both locking pins (Figure 17/1) simultaneously and push down ring (Figure 17/2).

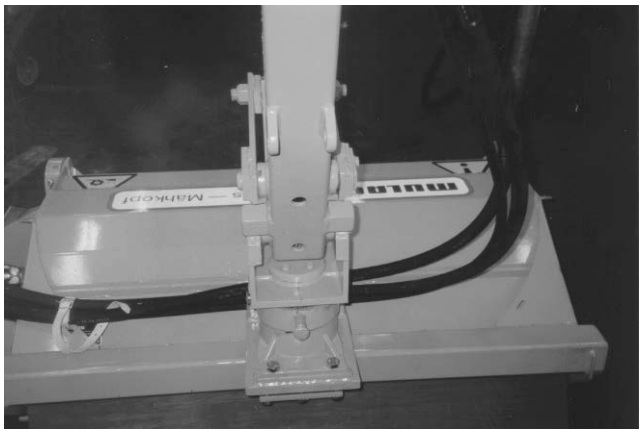


Figure 18

- Move boom and mower device to working position (Figure 18).
- Remove transport lock of mower device (see operating manual of device)

5.4 Changing and starting up the devices

5.4.1 Switching the device on / off

Switching on the device



- The mower devices may require special operating conditions – please refer to the relevant individual instruction manuals.
- Before switching on the mower device, check all fixing devices. Missing, damaged or worn devices must be replaced immediately by original spare parts from MULAG.
- The tools of the mower devices may take a while to slow down (see relevant instruction manual).
- If vibrations occur in the mower device during operation, switch off the device immediately, then investigate and eliminate the cause. If the error cannot be located and rectified, please contact the MULAG after-sales service department. Do not put the embankment mower back into operation in this case as this could cause further damage.
- The embankment mower may only be operated from the operating panel in the cab with the cab doors closed. The door window on the working side must be closed.

Mechanical activation of mower device by PTO shaft

- Switch on engine of carrier vehicle.
- Select front PTO shaft of carrier vehicle.
- Preselect speed setting according to carrier vehicle manual and section 2.2.
- Switch on PTO shaft according to carrier vehicle manual.



The mower device starts up when the PTO shaft drive is engaged.

- Select an engine speed at which the volumetric flow required for the mower device is obtained as shown in the volumetric flow diagram (see section 2.2).

Mechanical/electrical activation of mower device from operating panel



Check whether the mower device is switched off at the operating panel; if not, switch if off now.

- Switch on engine of carrier vehicle.
- Select front PTO shaft of carrier vehicle.
- Preselect speed setting according to carrier vehicle manual and section 2.2.
- Switch on PTO shaft according to carrier vehicle manual.
- Switch on service current supply via key switch in operating panel.
- Switch on mower device at operating panel.



Mower device starts up.

- Select an engine speed at which the volumetric flow required for the mower device is obtained as shown in the volumetric flow diagram (see section 2.2).
- **Switching off the mower device**



Allow the mower devices to slow down.

- Follow the same procedure as in section 5.4.1. Switching on mower device“ but in **reverse**.



Switching off procedure:
(The order described is very important and must absolutely be observed so that all operational data and parameters can be stored properly in the memory of the controller)

1. Switch-off all functions
2. Switch-off the operating panel
3. Switch-off the ignition of the carrier vehicle.

5.4.2 Dismounting the mower device

Always switch off the engine and hydraulic system before carrying out any work on the embankment mower.

- Deposit the mower device next to the carrier vehicle on properly consolidated horizontal ground (mount special holding fixtures as instructed in mower device manual if necessary).
- Disconnect hydraulic couplings and seal connections with protective caps. Place hydraulic hoses on the mower device or in the holding fixtures provided.

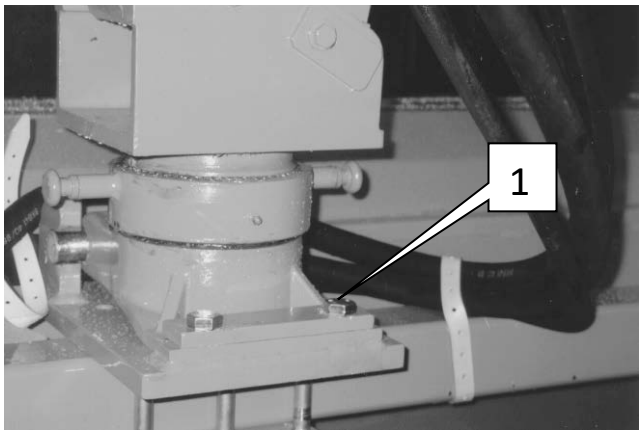


Figure 19

- Undo two screws (Figure 19/1) on Gearlock and remove clamping plate.

Make sure all functions in the operating panel are set to zero.

- Switch on operating panel.
- Raise Gearlock with the function “Swivel mower device“ and disengage from tool holding fixture with the function “Extend/bend boom“.

5.4.3 Mounting the mower device

Always switch off the engine and hydraulic system before carrying out any work on the embankment mower.

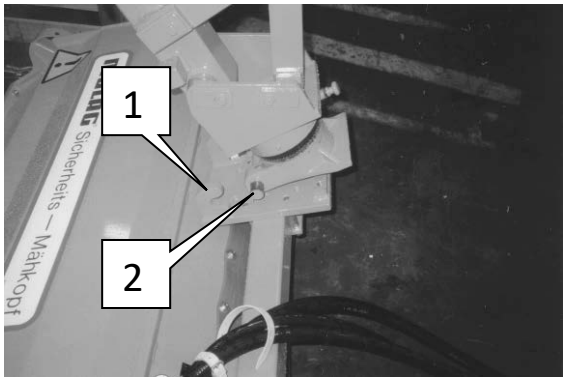


Figure 20

- Deposit the mower device by the side of the carrier vehicle.

or

- Approach the mower device sideways with the carrier vehicle.
- Extend boom over the mower device.
- Move Gearlock and mower device into position so that both lugs on the boom (Figure 20/1) can be inserted in the hooks of the tool holding fixture (Figure 20/2).

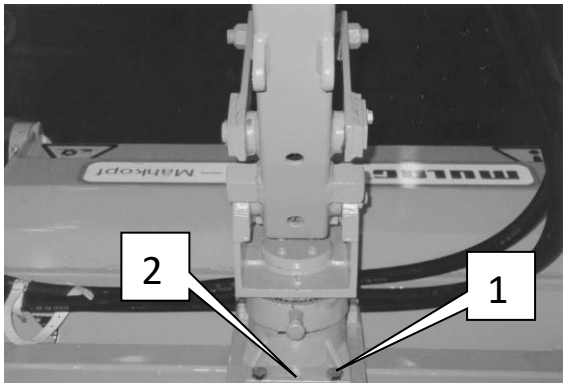


Figure 21

- Using the function “Swivel mower device“, rotate the Gearlock so that it rests fully on the tool holding fixture.
- Clamp tool holding fixture to the Gearlock with 2 screws (Figure 21/1) and the clamping plate (Figure 21/2).

- Lock Gearlock in place.

- Remove protective caps of hydraulic couplings. Connect hydraulic couplings according to colour coding. Pressure hoses are identified by the colour red.
- Dismount any special supporting fixtures according to the mower device manual.



- If cleanliness is not ensured, dirt can get into the hydraulic system via the couplings. In order to avoid damage to the entire hydraulic system, it is important to remove all dirt from the couplings.
- Check hydraulic couplings for firm seating and tighten securely.

5.5 Operating the embankment mower



- Danger from ejected objects (stones).
- The operator must ensure that there is nobody within the danger zone of the working attachment. Depending on the application, this safety zone increases to the side by the 9 m slewing range of the base unit.
- Danger from hydraulic liquids being sprayed under high pressure.
- Check connections and hydraulic hoses every day prior to starting work.
- Observe the operating regulations in the Operating Manual.
- Operators must be trained and qualified.
- If the combination mower hits a power cable, the driver/passenger must remain in the closed cab until the device is no longer touching the cable or until the cable has been de-energized.
- Only operate the working attachments if the safety equipment is in sound working order.
- Close cab doors and windows prior to operation.
- Do not touch hot surfaces when at operating temperature.
- Work can only be carried out by trained and authorized specialist personnel.



The operation is described in the separate operating instructions *(☞ m|control)*.

5.6 Preparing the carrier vehicle for travelling



- Always switch off the engine and hydraulic system before carrying out any work on the embankment mower.
- Make sure the mower device has come to a standstill.
- In case of equipments with Mähtronic function, always switch off the Mähtronic function before parking the machine, in order to avoid injuring people or damaging the boom system or the vehicle cab.

- Once the flail shaft / wash shaft has come to a standstill, swivel the mower device to the limit stop via the function **"Swivel/retract boom"** and lower it onto the ground.

- Switch off the engine of the carrier vehicle and remove the cuttings clinging to the mower device.



If any items are mounted on the roof, e.g. air conditioning system, raised roof, sunroof, etc., take care not to damage these when lowering the boom to the transport position.



Figure 22

- Switch on the hydraulic system.
- Raise the boom. After completing the work, rotate the mower device to the transport position. (Figure 22).



Figure 23

(FME 500)



Figure 24

(FME 400 with parking frame)



Figure 25

(FME 500/600 platform tray)



Figure 26

(FME 600) Transport support

5.6.1 Storage system: “parking frame“

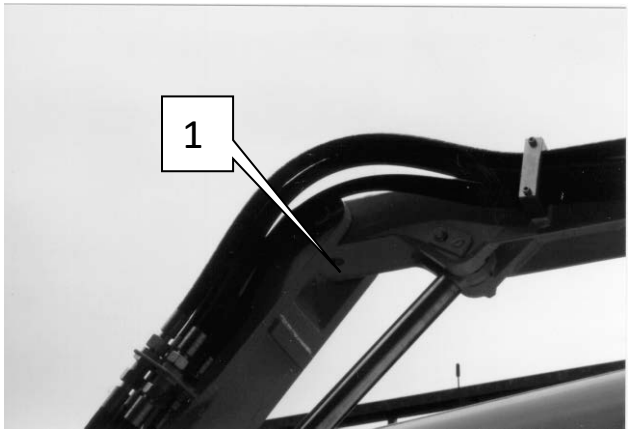


Figure 27

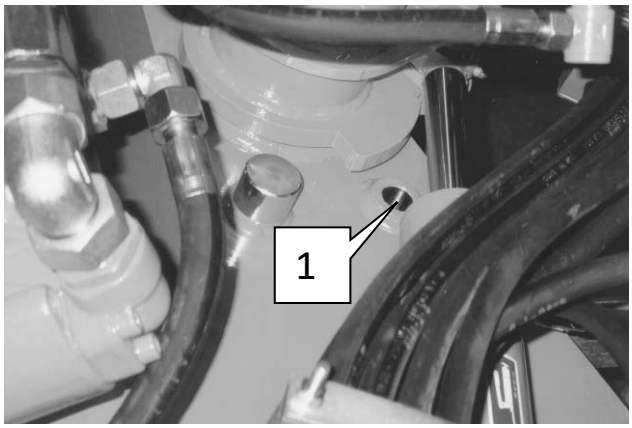
- FME 600T:Move the boom system to the centre of the vehicle.
- Swivel the mower device to the limit stop.
- Position boom 2 as shown (Figure 27).
- Make sure the lower edge of boom 2 (Figure 27/1) is parallel to the colour marking.

- Raise boom 1 vertically.
- Swivel boom system backwards over the loading surface until the 90° limit stop bolt is reached.
- Carefully lower boom 1 until boom 2 is resting on the parking frame.
- Once boom 2 is resting on the parking frame, lower the mower device to the transport position and lock boom 2 in the parking frame. (Figure 28)



Figure 28

- Lock boom to parking frame using safety hooks.



- Lock boom in place with a bolt (Figure 29/1) and switch off the hydraulic system.

Figure 29

- Close the ball cock for the "**Swivel mower device**" function.

The safety bolt (Figure 29/1) must not be inserted in connection with the parking frame during transportation. (keep safe in the vehicle tool set).

Transportation with the ball cock open may result in damage to the carrier vehicle and mower device.

5.6.2 Storage system platform tray and “transport support“

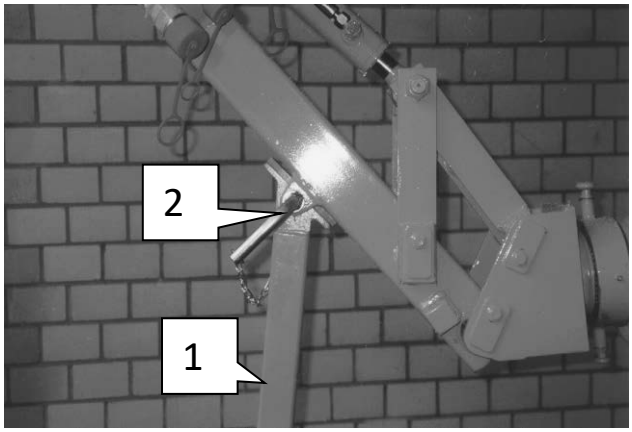


Figure 30

- Fix transport support (Figure 30/1) to boom with bolt (Figure 30/2) and secure with spring connector.
- Move FME 600 boom system to centre of vehicle.
- Swivel mower device to limit stop.
- Position boom 2 as shown in Figure 27.
- Move boom 1 to vertical position.
- Swivel boom system backwards over the loading surface until the 90° limit stop bolt is reached.
- Carefully lower boom 1 until the transport support or mower device is resting on the loading surface.
- Close ball cocks for the functions “**Lift boom**“, **Bend boom**“ and “**Swivel mower device**.



- If the mower device is stowed on the loading surface or transport support (only possible with the FME 500/600), the boom must be additionally secured for long-distance or transportation journeys with the second swivel limit stop bolt (Figure 36/3).
- The ball cocks must also be closed in case the carrier vehicle and mower device are damaged by the lowering of the booms.

- Check the carrier vehicle for road safety before commencing the journey.

5.7 Parking assistance kit



- Always switch off the engine and hydraulic system before carrying out any work on the embankment mower.
- Make sure the mowing device has come to a standstill.
- In order to prevent the mowing equipment from being damaged when swivelling the boom backwards in transport position, an electronic parking assistance is activated through a switch during the swivelling procedure.
- When activating the “parking assistance”, the execution speed of the functions “Lift/Lower”, “Bend boom”, “rotate boom” and “swivel cutter head” will reduced even if the operator has chosen the maximal speed on the joystick. The execution speed will be reduced as to ensure a safety parking position (transport position).
- The Mähtronic, floating position MKS, and arm relief system (AER) must always switched-off while parking the mowing equipment so that nobody can be injured or the boom or the cab of the carrier vehicle damaged.

- Start Motor.
- Adjust motor speed 800 - 1000 rpm.

- Switch on the hydraulic system.
- Switch on the operating panel.

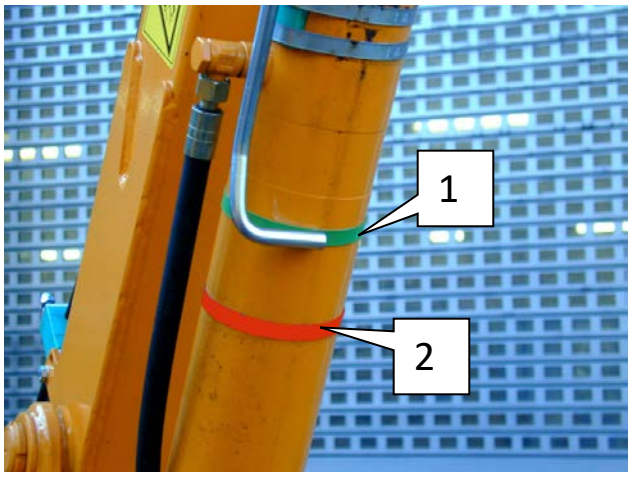


Figure 31

- Thanks to the operating panel, lift boom 1 until the indicator on the cylinder has left the red mark and reached the green one.

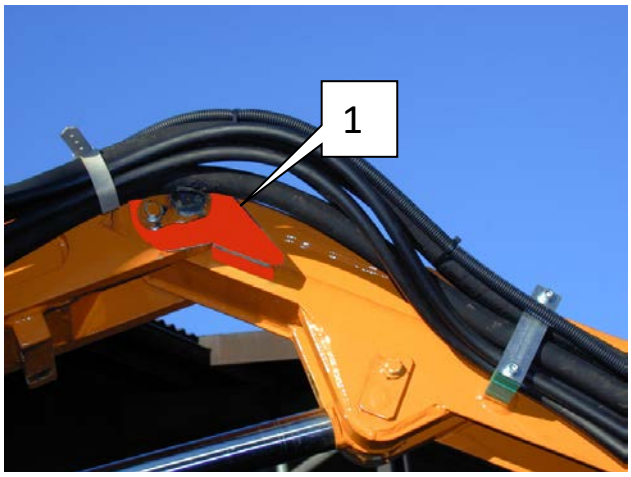


Figure 32

- Extend boom 2 until the angular red sheet (Figure 32/1) is parallel to the border of boom 2.
- When the green mark on boom 1 has been reached and the boom 2 has been placed correctly in relation to the red mark, a sufficient safety zone to the driver's is available for the swivelling procedure.



Figure 33

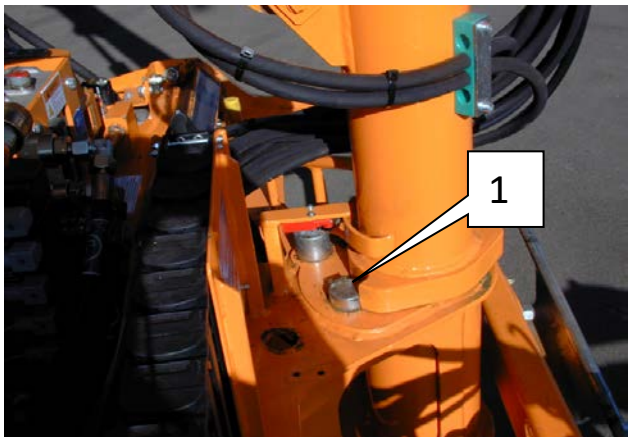
- In order to swivel to boom into the transport position, activate the function “rotate boom” to the swivelling limit.



Figure 34

- Place the lateral shift of the basic unit in the middle (see indication on the basic frame Figure 34/1)

- Carefully lower the boom system with working attachment until the indicator on the cylinder of the boom 1 coincides with the red mark.



- Lock the boom thanks to the locking pin (Figure 35/1) and switch-off the hydraulic system.

Figure 35

- Close the ball cock for the "**swivel working attachment**" function.

5.8 Changing over the mower device from right-hand to left-hand operation (or vice versa)

For the purposes of this description, the embankment mower is assumed to be in the transport position. The working side can be changed with or without the mower device.

- Park the carrier vehicle on an even, stable surface.
- Switch off the mower device and wait for it to come to a complete standstill.
- Always switch off the engine and hydraulic system before carrying out any work on the embankment mower.

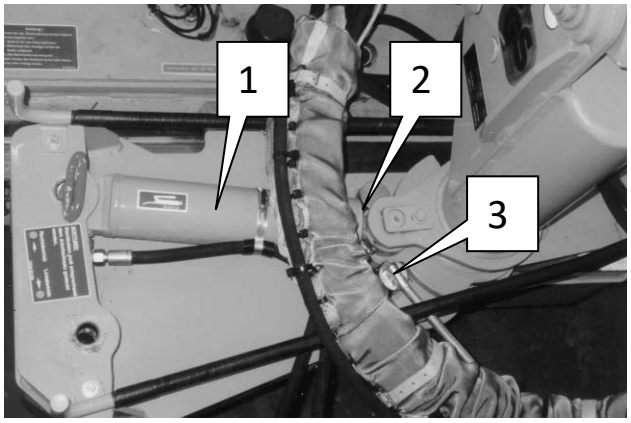
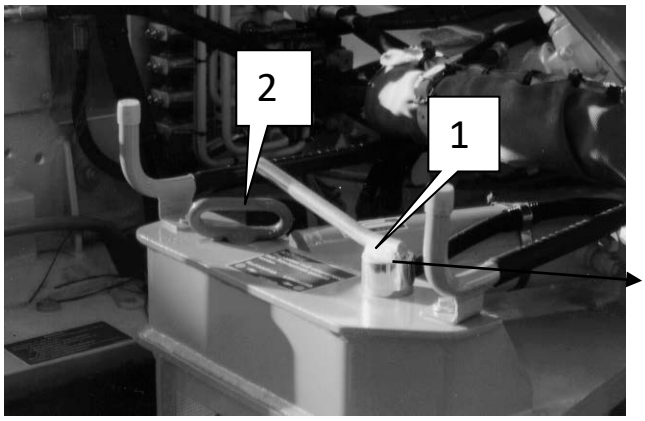


Figure 36

- Switch off the hydraulic system.
- Relieve pressure of swivel cylinder (Figure 36/1) by actuating the function**“ Swivel boom forward/backward“** with the emergency hand lever directly on the control block.
- Remove bolt (Figure 36/3) from parking position and insert bolt (Figure 36/2) in the hole (Figure 36/3).

To relieve the pressure, place the emergency hand lever on the **“Swivel boom“** segment of the valve block. After completing the work, keep the emergency hand lever safe in the vehicle tool set.



- Remove socket pin (Figure 37/2) and fix swivel cylinder at front insertion point (Figure 37/1) for left-hand operation.
  (Pin not shown in diagram)

Figure 37

- To put the embankment mower into operation, follow the instructions of section 5.1. in **reverse** order.
- To change over the embankment mower from left-hand to right-hand mowing, following the instructions of section 5.4. in **reverse** order.
- If a Mähtronic control unit is fitted, switch it over from right-hand to left-hand operation.

6 Dismounting the embankment mower



- Secure carrier vehicle to prevent it from starting up or rolling away accidentally.
- Set down embankment mower on a properly consolidated flat, horizontal surface.
- Always switch off the engine and hydraulic system before carrying out any work on the embankment mower.
- Keep a safe distance of 7 m from the embankment mower while the vehicle is moving away from the device in case it is knocked over.

- Reduce the spring deflection on one side of the front axle (lock MULAG torsion frame.)
- Release the boom from the parking frame, retract the blade head and extend the boom to the side in order to lower it onto the ground as a fourth support.
- FME 600: Move the boom system to the centre position.
- Switch off the engine of the carrier vehicle.
- Rotate the protective cover of the PTO shaft so that the locking pin of the cardan shaft can be seen. Push in the locking pin and remove the cardan shaft from the PTO shaft.
- Disconnect the control cable and service current cable and dismount the operating panel. Store the operating panel and loose cable somewhere dry.
- Push three supports (Figure 38/1) as far as possible into the locating fixtures provided (Figure 38) and lower the feet of the supports onto the ground by turning the crank handle.

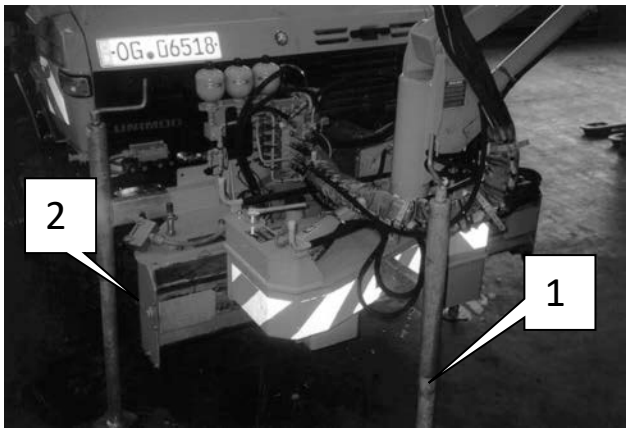


Figure 38

- Undo two collar screws (Figure 38/2) or release the Servolock.
- Lift the device above the three supports (Figure 38/1) using the crank handles until both engaging claws are disengaged from the carrier vehicle.

- Place emergency hand lever on the **"Lift/lower boom"** function on the control block.
- While raising the embankment mower, actuate the control lever for the **"Lift/lower boom"** function from time to time. This relieves the pressure of the lifting cylinder and allows the device to be raised more evenly.
- After completing the job, remove the emergency hand lever and keep it safe in the vehicle tool set.

- Disconnect the hydraulic couplings and seal the connections with protective caps.
- Connect the separate return conductor of the control system to the leakage oil tank.

The device control system must not be actuated again after disconnecting the hydraulic hose-pipes from the hydraulic system, otherwise the control unit may be damaged.

- Start up the engine of the carrier vehicle.
- Carefully move back the carrier vehicle.
- Dismount the parking frame and ballast weight.
- To dismount the parking frame and ballast weight, follow the assembly procedure (see section 4.4) in reverse order.

7 Servicing and maintenance



- Observe safety instructions in chapter 1.
- Always switch off the engine and hydraulic system before carrying out any work on the embankment mower.
- Repairs may only be performed by suitably authorised specialists.
- The specified intervals for carrying out the individual maintenance jobs must be strictly adhered to.
- In the case of screw joints with different property classes, choose the tightening torque of the lowest property class.

7.1 Torque specifications

Application

MULAG determine the tightening torques for the screwing connections depending on the steadiness, the measure and the surface covering of the screws. Should other tightening torques be necessary for special applications they are notified in the operating manual.

Description of the terms used in the following charts

Gewindepaarung (thread couple): Here the type of surface coating of screws and bolts is fixed.

Schw (black): The thread of the screw or bolt is black, lightly lubricated or not.
The thread of the screw or bolt is chrome (galvanised), lightly lubricated or not.

Verz: The thread of the screw or bolt is chrome (galvanised), lightly lubricated or not.

Stoppmutter: Self-locking bolt, chrome-covered

µ: reference value of the total driving value of the thread couple.

When using MoS_2-Pastes as lubricants, the tightening torques for self-locking nut must be reduced of 10%.

Tightening torque for screw joints: metric thread (all data are given in [Nm])									
	8.8			10.9			12.9		
Thread	Black Black µ=0,14	Galvanized Galvanized Black µ=0,1	Self-locking nut galvanized	Black Black µ=0,14	Galvanized Galvanized Black µ=0,1	Self-locking nut galvanized	Black Black µ=0,14	Galvanized Galvanized Black µ=0,1	Self-locking nut galvanized
M 4	3	2,4	3	4	4	4	5	4	4
M 5	6	4,8	5	9	7	8	10	8	9
M 6	10	8,3	9	14	12	13	17	14	15
M 8	25	20	23	35	30	32	41	35	37
M 10	49	40	44	69	59	62	83	69	75
M 12	86	69	77	120	100	108	145	120	131
M 14	135	110	122	190	160	171	230	190	207
M 16	210	170	189	295	250	266	355	290	320
M 18	290	245	261	405	345	365	485	405	437
M 20	410	340	369	580	490	522	690	570	621
M 22	550	460	495	780	660	702	930	770	837
M 24	710	590	639	1000	840	900	1200	980	1080
M 27	1050	870	945	1500	1250	1350	1800	1450	1620
M 30	1450	1200	1305	2000	1700	1800	2400	1950	2160

Tightening torque for screw joints: metric thread (all data are given in [Nm])						
	8.8		10.9		12.9	
Thread	Black Black μ=0,14	Galvanized Galvanized Black μ=0,1	Black Black μ=0,14	Galvanized Galvanized Black μ=0,1	Black Black μ=0,14	Galvanized Galvanized Black μ=0,1
M 8x1	27	22	38	32	45	37
M 10X1,25	52	42	73	62	88	72
M 12x1,25	95	75	135	110	160	130
M 12x1,5	90	72	125	105	150	125
M 14x1,5	150	120	210	175	250	200
M 16x1,5	225	180	315	265	380	310
M 18x1,5	325	270	460	385	550	450
M 20x1,5	460	375	640	530	770	620
M 22x1,5	610	500	860	710	1050	830
M 24x2	780	630	1100	900	1300	1050
M 27x2	1150	920	1600	1300	1950	1550
M 30x2	1600	1300	2250	1850	2700	2150

Maximum tightening torque $M_{A\,max}$ Nm Hexagonal and cylinder screws with deep heads												
Thread	8.8	Inox A2-70 A4-70	8.8	Inox A2-70 A4-70	10.9	12.9	10.9-	Inox A2-70 A4-70	10.9	Inox A2-70 A4-70	45 H	Inox A2-70 A4-70
M 3			1	0,6			1	0,5	1	0.5	0,5	0,2
M 4	2	1	2	1,2	2		2	1	2	1	1	0,5
M 5	6	4	4	2,5	5	4	5	2,5	4	2	3	1,5
M 6	9	5	8	5	5	9	9	4.5	8	4	5	2,5
M 8	20	12	12	7	10	25	15	8	12	6	10	5
M10	40	24	35	21	18	40	40	20	30	15	20	10
M12	65	40	50	30		70	65	33	60	30	45	22
M14	110	66					100	50			46	22
M16	180	110	110	66		200	110	55			90	45
M16											140	70
M20	260	170	200	120		400	150	75			140	70
M 22											220	110
M2A			390	235			400	200			220	110

7.2 Care

	When using steam or high pressure cleaners, do not aim the water jet at electrical or hydraulic equipment. This poses the risk of water penetrating and destroying the electrical or hydraulic systems.

	Activity	**Care interval**
Embankment mower	• Remove dirt and remnants of grass and undergrowth. • Remove caked-on grass from mower attachment.	Daily

7.3 Inspection

	Activity	Inspection interval
Embankment mower	• Inspect for visible damage or defaults; if damage or defects are discovered, find out the cause and take effective corrective action.	Daily
	• Check holding fixture for adjusting hydraulic pump.	Once a week
	• Check for disorder (e.g. loss of speed or power).	
Hydraulic conductors	• Check for leakages and seal immediately.	Daily
Protective devices (in the region of the cardan shaft and lateral shift)	• Check for wear, completeness and firm seating; replace worn or missing parts immediately and secure any loose parts.	Daily before starting any work
Pivot	• Check limit stop elements at the pivot bearing for tight fit.	Once a month
Electrical system	• Check entire electrical system for dirt.	Once a month
	• Check lines and cables for sound condition and fastening (no mechanical damage or discolouring of insulation).	Once a month

	Activity	Inspection interval
	• Check screw and plug connections for firm seating.	Once a month
	• Check operating and indicating elements, key switches and indicator lights for sound condition (no mechanical damage) and working order.	Daily before starting any work
	• Make sure electrical control box shows no signs of mechanical damage and is tightly sealed.	Monthly
Hydraulic oil	• Check filling level in inspection glass (if glass is damaged or temperature display illegible, replace oil level/temperature indicator). Do not mix different hydraulic oils.	Daily
	• Check temperature in inspection glass. If the temperature exceeds 85°C, stop using the embankment mower and check the pressure limiting valve/pilot valve.	Several times a day during work breaks

	Activity	**Inspection interval**
Diaphragm accumulator	• Check pressure setting and test for leaks	Every 1000 operating hours. Must be carried out by specialists.
Chain tension of cross function	• Traverse boom system to the left limit position (towards the hydraulic motor) and tension chain using the clamping screws so that the upper strand of the chain has approx. 10 mm slack at the centre due to its own weight.	Once a week
Hose connections to carrier vehicle and working attachments	• Check for firm seating/externally visible damage.	Daily before starting work and after changing attachments
Piston rod of the hydraulic cylinder	• Grease	After each mowing season and for storing

7.4 Maintenance work

7.4.1 Maintenance and lubrication schedule

For process materials, see section7.4.8

Activity	Activity described in section	Maintenance intervals				
		10 (daily)	**50** (weekly)	**200** (month-ly)	**1000** (yearly)	
Re-grease boom joints, cylinder bearing points and slewing ring [1)]			●			
Check screw and hydraulic connections, tighten if necessary		○	●			
Change suction filter element	7.4.2		●			1 x yearly
Change pressure filter element	7.4.3		●			1 x yearly
Change oil in hydraulic system 2)	7.4.4				●	After one year at the latest
Oil change on the hydraulic unit [2)] (filled with PANOLIN HLP SYNTH 46)	When using PANOLIN HLP SYNTH in accordance with the warranty conditions, it is possible to reach oil change intervals which are at least 5 times longer (depending on the use of the machine) than the oil change intervals generally recommended for mineral oil based hydraulic fluids.					5000 h
Check pressure of hydraulic system 2)	7.4.5					
Re-grease exposed grease lubrication points 4)		●				Every time you clean the device
Check storage supports for damage 5)						

Activity	Activity described in section	Maintenance intervals				
		10 (daily)	**50** (weekly)	**200** (month-ly)	**1000** (yearly)	
Re-grease PTO shaft and claws of Centaflex coupling 6)	7.4.6					After one year at the latest
Check diaphragm accumulator according to hydraulic circuit diagram 7)	7.4.7					
Re-grease Gearlock			●			
Re-grease hydraulic rotating device			●			
Re-grease piston rods of the hydraulic cylinders					●	Before stowage
Re-grease lubricating nipples of hydraulic implement turning device [8)]			●			
Check limit stop elements at the pivot for tight fit.				●		

○ = First time ● = All subsequent times

1) Re-grease lubricating nipples with a grease gun (max. 2 shots).
2) Once the embankment mower has been set down on the supports, change the oil in the hydraulic system.
3) See the instructions in the section quoted adjacently.
4) Exposed grease lubrication points include the guideways of the cross mechanism
5) Before dismounting the embankment mower from the carrier vehicle.
6) When mounting the pump drive.
7) In case of malfunctions.
8) Lubricate with water-repellent multi-purpose grease OKS 420.

7.4.2 Changing the suction filter element

- Do not clean filter element.
- Dispose of used filter elements and any oil escaping during the filter change properly in accordance with environmental protection regulations.
- Keep suction filters free of dust and dirt.

The suction filter is equipped to ensure that no more than 2 l of oil can escape when changing the filter element. The stored oil can be left in the tank during the filter change.

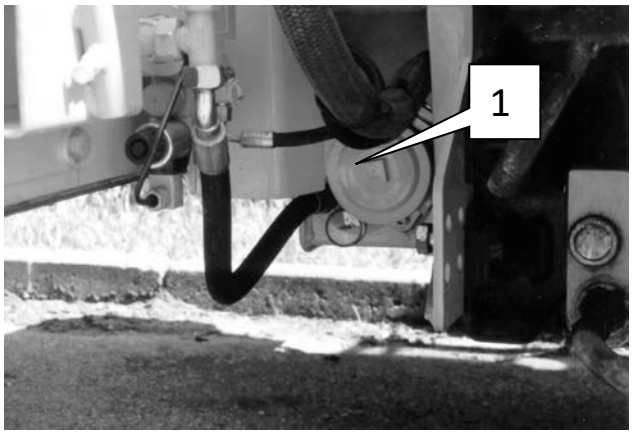


Figure 39

- Unscrew suction filter cover (Figure 39/1).
- Take out filter element and replace with a new one.
- Re-mount suction filter cover and fasten.

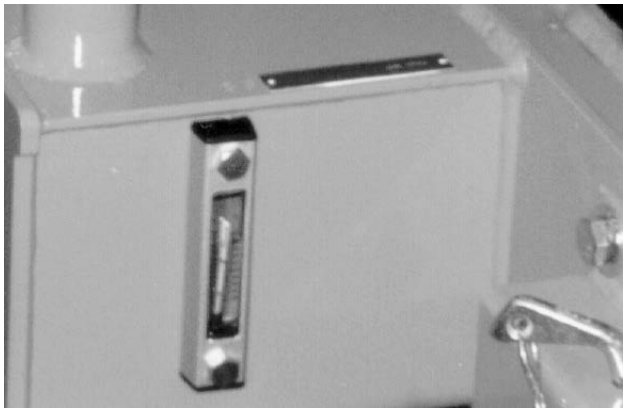


Figure 40

- Check filling level of hydraulic oil in inspection glass and top up through the filler neck if necessary. Make sure there is no dirt in the region of the filler neck.

7.4.3 Changing the pressure filter element

- Do not clean filter element.
- Dispose of used filter elements and any oil escaping during the filter change properly in accordance with environmental protection regulations.
- Keep pressure filters free of dust and dirt.

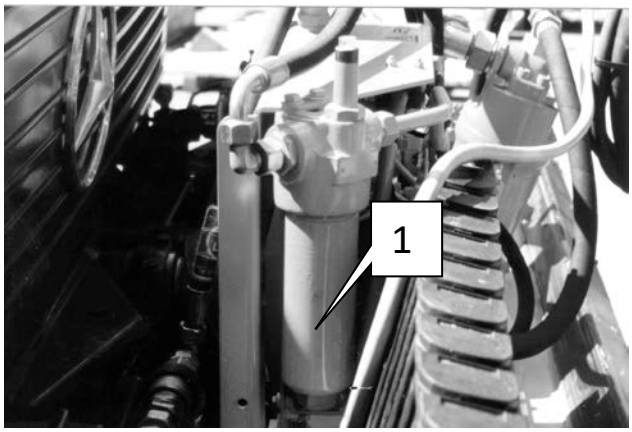


Figure 41

- Unscrew filter hood (Figure 41/1) at the bottom
- Change filter element.
- Screw filter hood (Figure 41/1) back on from the bottom.

7.4.4 Changing the oil in the hydraulic system

- Work on the hydraulic system may only be performed by suitably authorised specialists.
- Do not allow hydraulic oil to seep into the ground.
- Dispose of waste oil properly according to environmental protection regulations.

- Before changing the oil, it is essential to check the miscibility of the hydraulic oils.
- Should the oil specification change, it is necessary to replace the entire volume of oil (including the oil in the consumers and hoses).

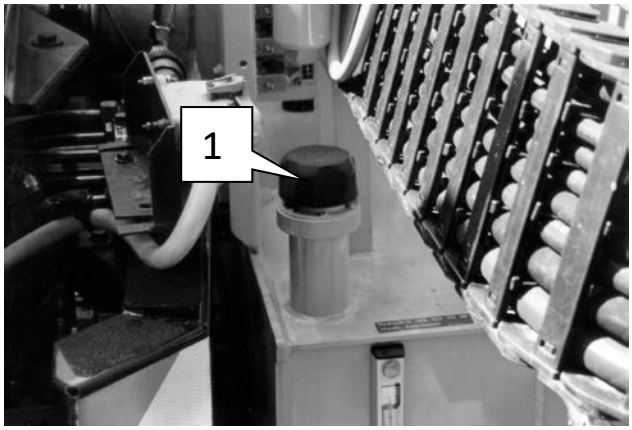


Figure 42

- Place the device over a collecting tank with a capacity of over 60l.
- Remove cover (Figure 42/19) from oil filler neck.
- Undo oil drain plug (Figure 42/1) in tank and drain off the waste oil.

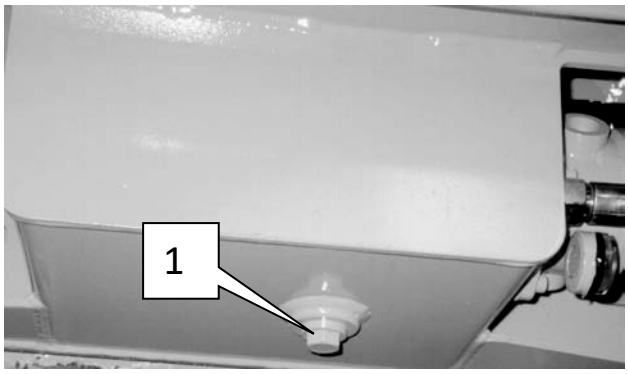


Figure 43

- Screw in oil drain plug with new copper sealing ring.
- Pour hydraulic oil in through filler neck (for oil type, see list of process materials and hydraulic oil recommendation in section 7.4.8)
- Remove any escaping oil immediately from enamelled surfaces – once resinified, hydraulic oil is hard to shift.

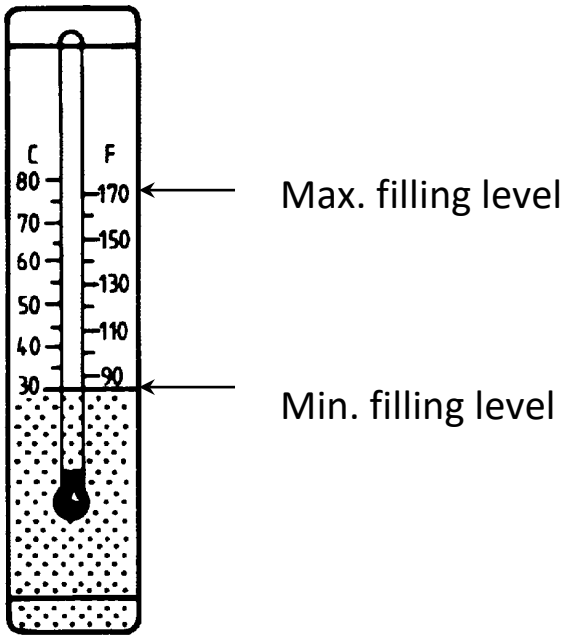


Figure 44

- Check filling level in inspection glass.
- Test-run device and check for leaks.
- Check filling level in inspection glass (Figure 44) again.
- Mount cover on filler neck and seal tight.

7.4.5 Checking the pressure of the hydraulic system

- The pressure in both oil circuits of the embankment mower can be checked with a pressure gauge. The measuring points are marked on the hydraulic diagram (see section 9.1) and can be easily identified on the device.
- measuring connections correspond to the system “Hydraulic engineering, screw series 1620“. The setpoint values can be found in section 9.1.

Oil circuit	Measuring point
Drive of mower attachment P1	On pressure limiting valve **or** on electric blade shaft pilot valve.
Boom functions	(Supplied with oil by carrier vehicle).

7.4.6 Lubrication of the cardan shaft

It is essential to lubricate the double joints before putting the device into operation for the first time. The lubrication intervals indicated in the diagram (Figure 45) must be observed.

☞ Failure to observe the prescribed lubrication intervals will greatly reduce the lifetime of the mower drive.

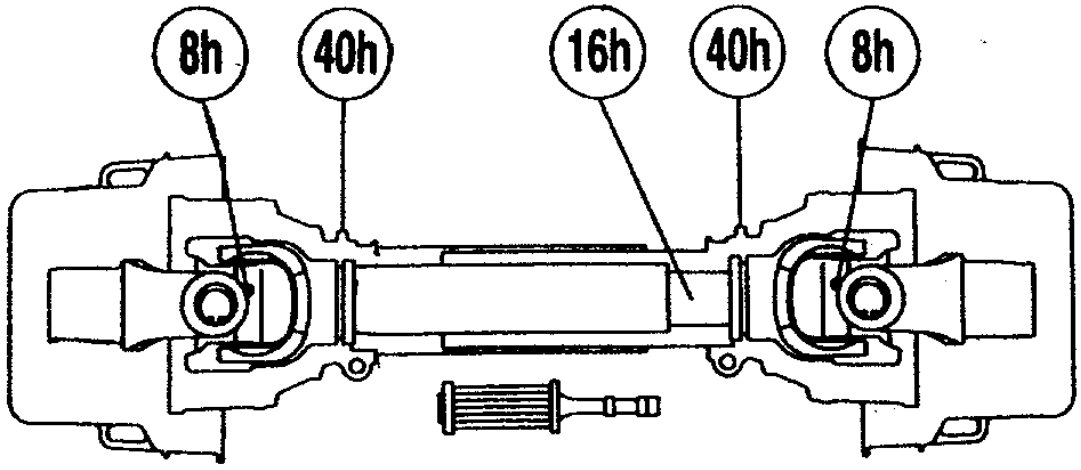


Figure 45

7.4.7 Controlling and setting the diaphragm accumulator

To control and set pressure it is necessary to use an accumulator filler of the company Hydak (UCA 01), a leak detecting spray and a bottle of nitrogen. These control and setting works can also be carried out by BOSCH as well as one of the Hydak service branches. The diaphragm accumulator are filled with inert gas 2,5 N2 and (nitrogen) and are in charge of 2 functions in MULAG embankment mowers :

1. Damping shocks due to external forces.
2. Refilling of trapped oil.

Both features are ensured by the gas cushion. Should the gas pressure bias be lost (because of leakage) the accumulator would be ineffective and it might lead to damages at the embankment mower.

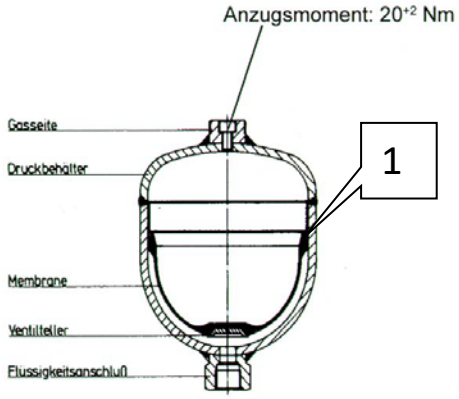


Figure 46

- The diaphragm accumulators (Figure 46/1) should be controlled once a year or after 1000 operating hours. The inspection procedure consists in checking the condition (twisted Hoses, deformations etc.) and controlling pressure via an accumulator filler.

The accumulator filler will be screwed on top of the diaphragm accumulator and it is then possible to read the accumulator pressure thanks to a manometer (For Inflation pressure see hydraulic circuit diagram Section. 9.1).
If necessary, the inflation pressure can be increased via the inflation tube at the accumulator filler and a bottle of inert gas - while controlling with a manometer.

Always observe following recommendations when setting pressure :

1. Once the nominal value has been reached for the first time, WAIT ONE MINUTE (compensation of temperature), then rectify the pressure and tighten the screw with predetermined tightening torque.



   Reason : When draining the gas cools extremely. The manometer of the accumulator filler does not indicate the pressure by ambient temperature but a lower one.
2. After setting the pressure check for leakage at the filler screw with a leak detector (spray).
3. The nominal values are listed in the hydraulic circuit diagram.

7.4.8 Process materials

Used for	Quantity	Lubricant specification	Viscosity
Hydraulic system	Approx. 60 l	Hydraulic oil HLP 46 ISO – mineral oil -	ISO VG 46
		Synthetic ester-based hydraulic fluid	VG 46 HEES 46
		Hydraulic liquid PANOLIN HLP SYNTH	HLP 46 SYNTH
Lubricating nipples / exposed grease lubrication points		Multi-purpose lithium base saponified grease K 2 K DIN 51825	
PTO shaft profile		Assembly paste for preventing fretting rust.	
hydraulic implement turning device		multi-purpose grease OKS 420	

*electrical conductivity > 300 pS/m

- This MULAG embankment mower works reliably and economically with lubricative pressure media over a wide viscosity range. For normal operation, we recommend one of the hydraulic oils specified by DIN 51524 Part 2 ISO VG 46.
- For applications at low outdoor temperatures, there are special commercially available hydraulic oils (please check miscibility).
- When using synthetic oils (ester based oils) always observe the manufacturer’s guidelines concerning the miscibility with mineral oils or with other ester based oils.
- Before filling the hydraulic system with non-approved hydraulic oils, please consult the MULAG after-sales service department.

7.5 Repairs



- Always switch off the engine and hydraulic system before carrying out any work on the embankment mower.
- Repair work may only be performed by suitably authorised specialists.
- If you are unable to repair a fault, please contact our after-sales service department.

7.5.1 Welding repairs on the embankment mower



- An adequately skilled person must select the welding method and fillers.
- Our boom systems are partially manufactured with high-strength fine grain structural steels. Only companies benefiting from a qualification in welding high-strength fine-grain structural steels are authorised to carry out welding works on these steels.
- Always preheat transducing parts to 150° prior to welding them.

- Always remove plug connections (Figure 48), (Figure 49), (Figure 50) before performing any welding work on the embankment mower or carrier vehicle.
- If the embankment mower is mounted on the carrier vehicle, the battery of the vehicle must be disconnected (see carrier vehicle manual for details).

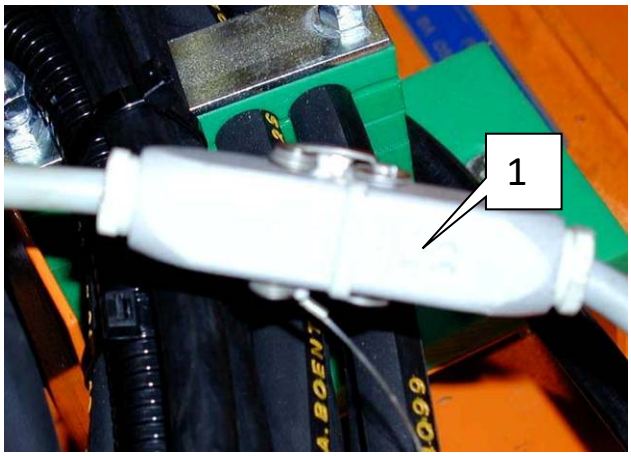


Figure 47

- Disconnect and remove bare plug from operating panel on electrical control cabinet (Figure 47/1).

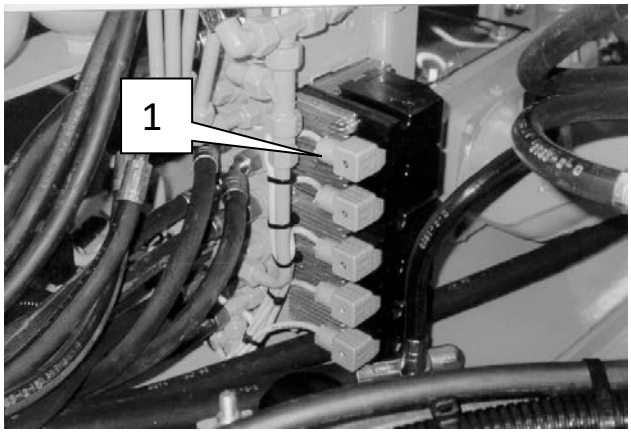


Figure 48

- Disconnect control cable from electrical control box.
- Disconnect cube plug (Figure 48/1) from control valves.



Figure 49

- Disconnect control cable from connecting point.

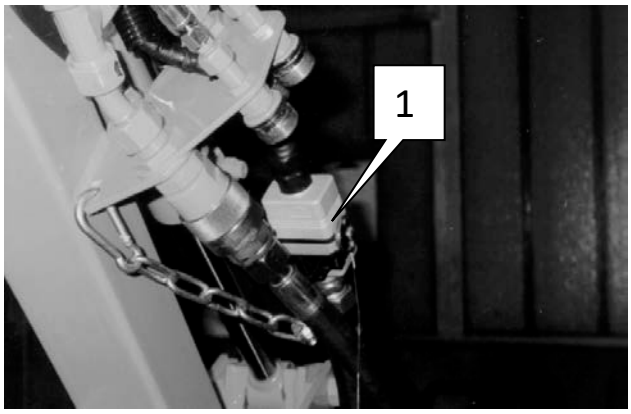


Figure 50

- Disconnect control cable from connecting point of boom. (Figure 50/1)

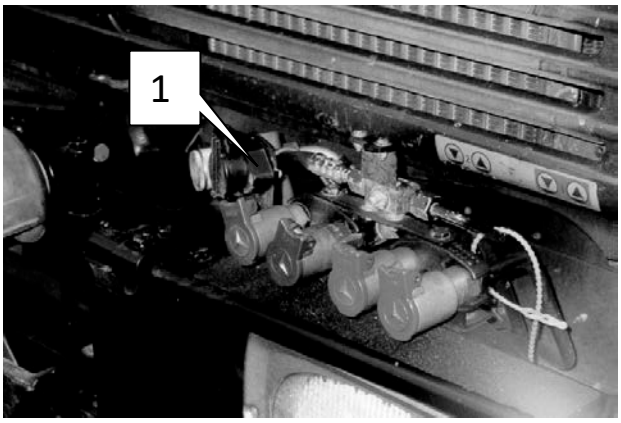


Figure 51

- Disconnect plug connection from service current socket. (Figure 51/1).

8 Faults / errors



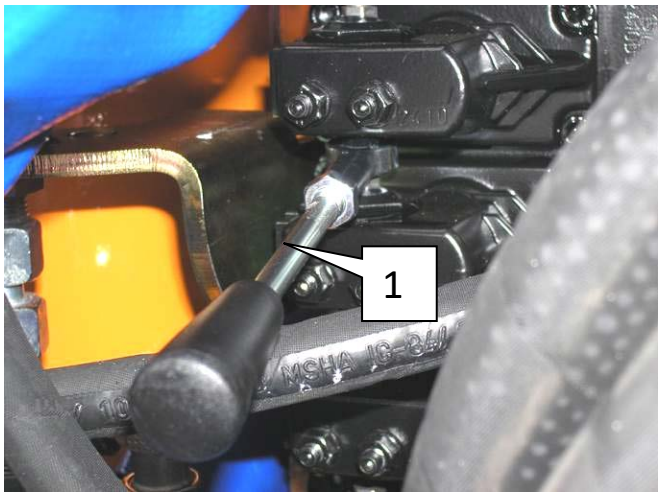
- Only carry out work on the hydraulic system when the electrical and hydraulic system are deactivated.
- Switch off the vehicle and take off the ignition key.
- Work may only be carried out by trained and authorised qualified personnel.

8.1 Emergency manual control



- During activation of a function, the operator remains in the danger zone.
- Remaining in the danger zone of the function being activated is unavoidable. Extreme caution and continuous monitoring of the boom movements and particularly the sideways movement is therefore required.

- If the boom mower can no longer be operated via the operating panel, it can be operated via an emergency hand lever (tool set) instead.
- Do not operate the control block with the emergency hand lever until the operating panel has been disconnected from the electrical system of the vehicle.
- Reduce the engine speed of the trailer vehicle to standby gas operation (approx. 700 rpm) in order to drastically slow the movement of the boom and the sideways movement, which is required for safety reasons.



- (Figure 52/1) shows the emergency hand lever on an operating element of the control block.

Figure 52

8.2 Error table

☞ In the event of malfunctions or errors that are not described in the table below or which cannot be rectified through referring to the table, please contact our Customer Service.

Fault / error	Cause	Remedy
Boom cannot be operated	Power supply switched off	Switch on power supply (key switch)
	Power supply connection not plugged in	Plug in power supply connection
	Hydraulic system of carrier vehicle switched off	Switch on hydraulic system
	Electrical control system defective	Check control system (after-sales service)
	Hydraulic connection to carrier vehicle interrupted	Check quick coupling and hose-pipes
	Pressure limiting valve of control block dirty	Clean valve
Mower device performance deteriorating	Coupling connection on boom loose	Tighten screw couplings and/or check for externally visible damage
	Pump or oil motor defective	Check operation of pump or oil motor, consult after-sales service

Fault / error	Cause	Remedy
	Pressure limiting valve dirty	Clean valve
Mower device motor leaking at rotary shaft seal	Sealing ring defective	Replace sealing ring
	Oil recirculation or leakage oil hose jammed	Check hoses and if necessary undo connections, re-install hoses and re-tighten connections
	Quick coupling defective	Replace coupling
Boom swings back during mowing	Swivel cylinder defective	Seal swivel cylinder Consult after-sales service
	Pressure limiting valve dirty or defective	Clean valve or replace if necessary
Side traversing mechanism cannot be operated	Guides dirty or poorly lubricated	Clean and lubricate guides
	Sliding segments worn Too much guide play	Replace sliding segments with new ones Readjust guide play above the lower sliding elements
Boom cannot be lowered or falls down when lowered	Lowering throttle damaged or displaced	Shut down device, notify MULAG after-sales service
Boom system dropping	System-related leakage of control block	Retrofit seat valves

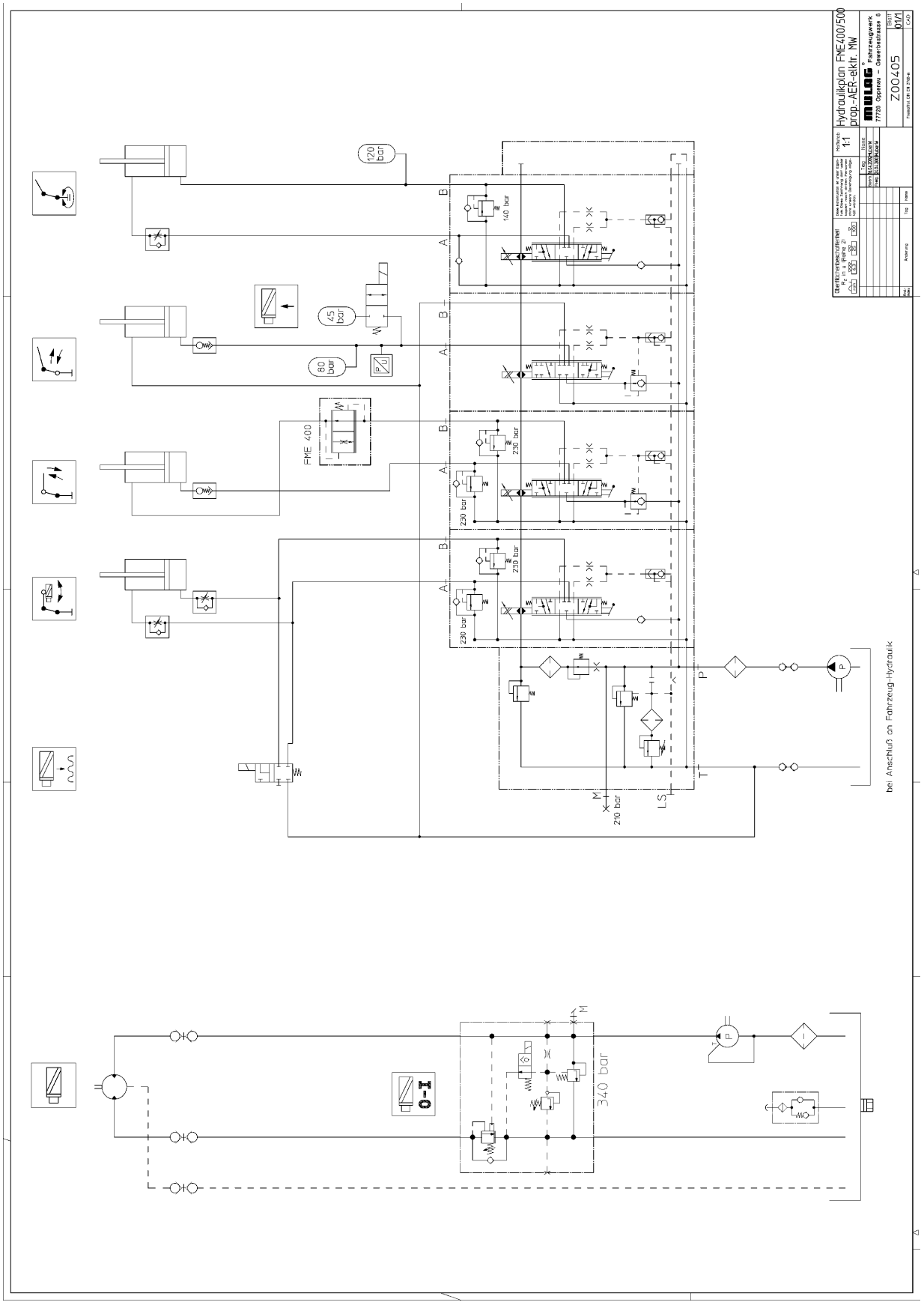
Fault / error	Cause	Remedy
Foaming of hydraulic oil	Leak in suction hose	Seal or replace suction hose
“Bend boom“ function not working	Lowering throttle or conductors damaged or displaced	Shut down device, notify MULAG after-sales service

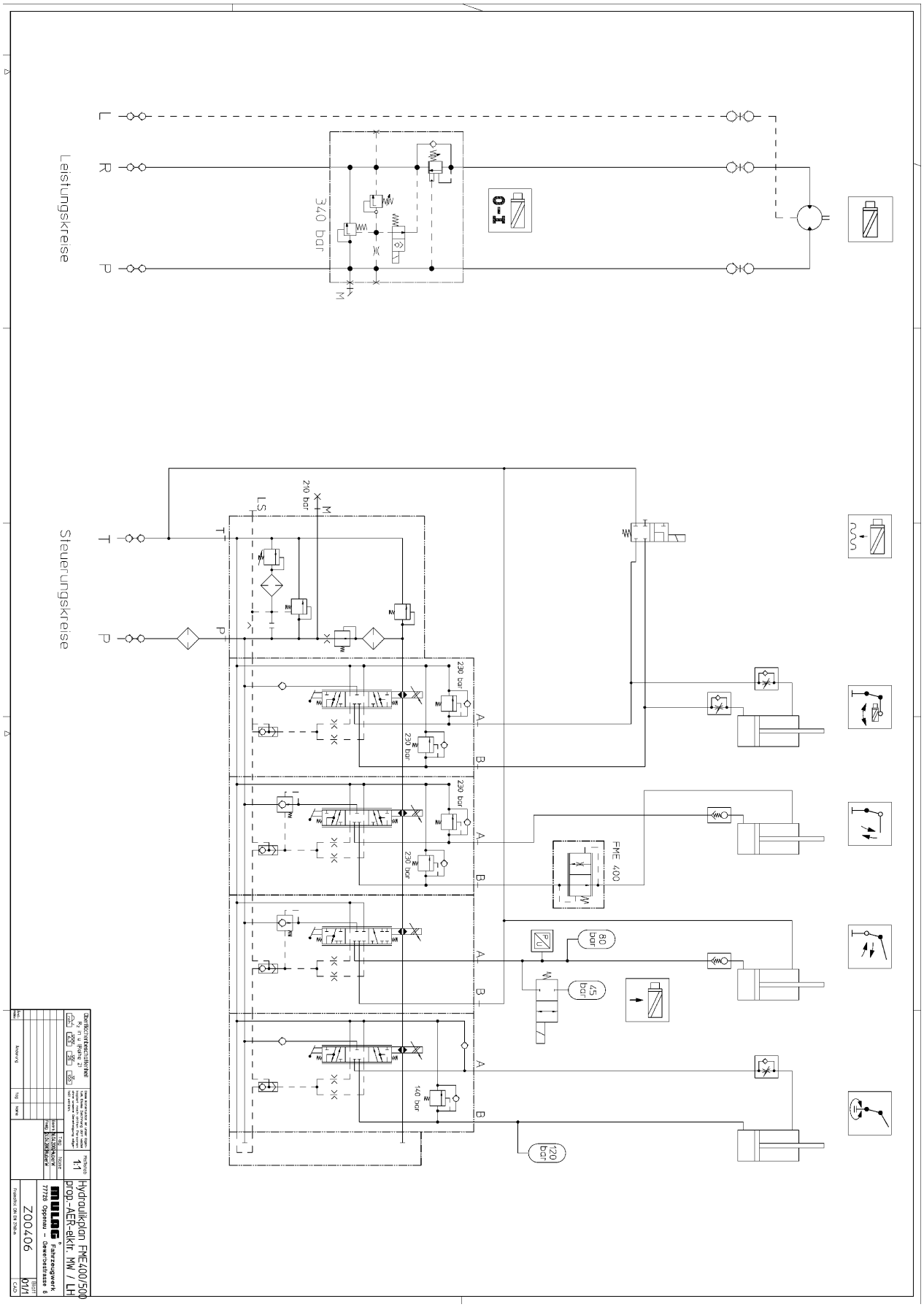
8.3 Fault detection via diagnostic mode



To obtain some information on the fault detection via diagnostic mode please consult the *Workshop Manual*.
The *Workshop Manual* is always delivered together with the equipment.

9 Hydraulic circuit diagram




Leistungskreise
L
R
P
340 bar
M
I-0
Steuerungskreise
T
P
LS
210 bar
230 bar
230 bar
230 bar
230 bar
FME 400
80 bar
45 bar
140 bar
120 bar
A
B
Hydraulikplan FME400/500
prop.-AER-elktr. MW / LH
MULAG Fahrzeugwerk
77728 Oppenau – Gewerbestrasse 6
Z00406
01/1

9.1 Fuse configuration

The machine specific fuse location is to be found in the circuit diagrams of the spare parts list

10 Optional extras

10.1 Hydraulic turning gear

- Depending on the attachment, the use of the hydraulic turning gear changes the direction of the ejected matter (e.g. dirt), resulting in a risk of injury.
- Always switch off the engine and relieve the hydraulic pressure before carrying out any work on the embankment mower and attachment.

- The hydraulic turning gear is operated via the additional function of the operating panel (see chapter 3).
- Lay the hydraulic hoses in such a way as to prevent them from being damaged during operation.
- Lubricate the implement turning device weekly or after every 50 operating hours with water-repellent multi-purpose grease (OKS 420).

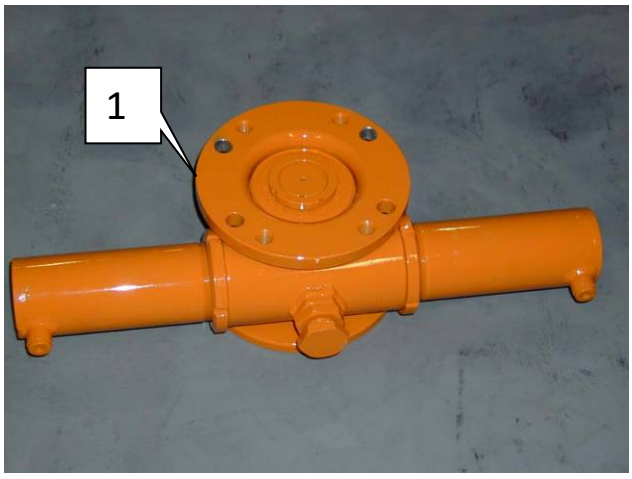


Figure 53

- If the embankment mower is equipped with an optional slewing device for the working attachments (Figure 53/1) the fixture of mowing head can be turned into the required position from the driver's cab (thanks to the additional function at the control panel described in section 3)

- The gearlock can remain locked permanently and it is not necessary to release it by hand before turning the working attachment.

10.1.1 Converting swivel drive (Item No.: 176546) to right or left-hand operation when fitted

- Observe the safety instructions (see Chapter 1 of the Operating Manual for the attachment)
- Always switch off the engine and relieve the hydraulic pressure before carrying out any work on the attachment.
- Do not touch attachments while in motion. Wait until the cutter has come to a complete stop.
- Maintenance work may only be carried out by appropriately authorized specialists.
- In the case of threaded joints with different strength classes, choose the tightening torque of the lowest strength class.

10.1.1.1 Setting up

Start vehicle and mowing device as per the Operating Instruction, bring boom system into working position and dismount attachment.

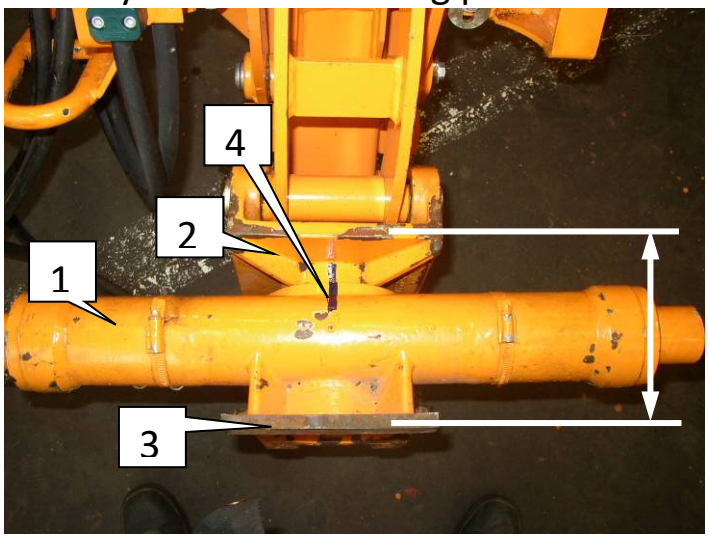


Figure 54

- Affix marking (Figure 54/4) for monitoring purposes.
- Rotate swivel drive (Figure 54/1) so that the surface (Figure 54/3) of the connecting part is parallel with the surface (Figure 54/2) of the base carrier.

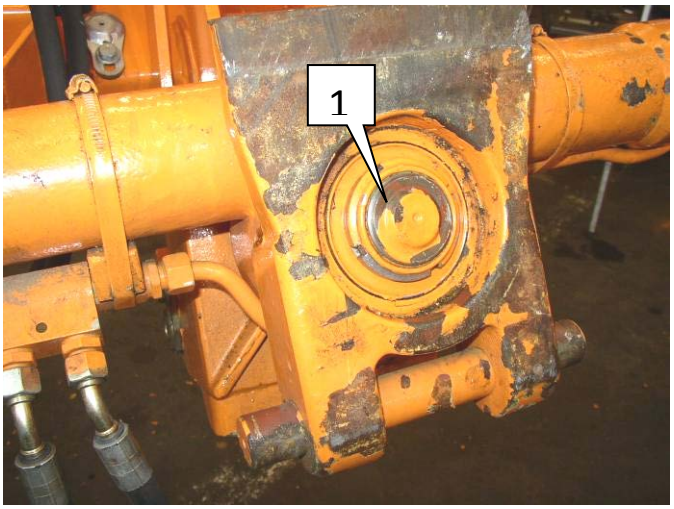


Figure 55

- Move boom system up to the groove nut (Figure 55/1).

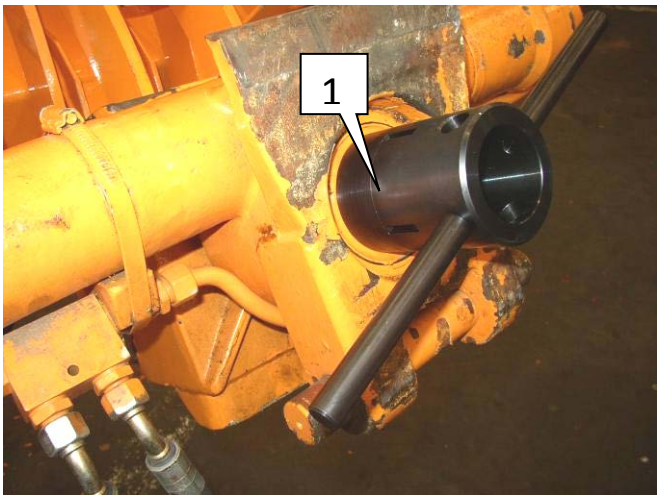


Figure 56

- Use a suitable wrench to loosen the M 50 x 1.5 groove nut.
  A pin wrench (Figure 56/1, Item no.: 173640) can be ordered from MULAG.

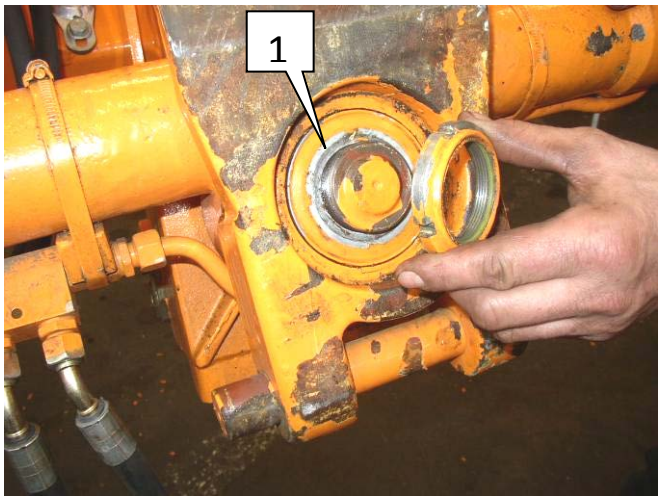


Figure 57



Figure 58

- Remove groove nut, Nilos ring (Figure 57/1) and tapered roller bearings.
- Using a plastic hammer, lightly tap the swivel drive until the bottom part is detached from the tapered roller bearings.

Take care to ensure cleanliness!
Toothed wheels must not become contaminated.



Figure 59

- The mounting base must be fitted offset by 90°, depending on use for right or left-hand operation.

10.1.1.2 Mounting for left-hand operation



Figure 60

- Mount 90° (or less, depending on the tooth position) rotated to the **right**.

> After rotation, the angle may differ depending on the original position.

- Mount tapered roller bearings, Nilos ring and groove nuts and tighten until the tapered roller bearings are firmly seated.

> The groove nut (Item no.: 165563) is self-locking and must be replaced.

- After conversion, check function and mount the attachment (cutter head) as per the Operating Manual.
- Lubricate the swivel drive.

10.1.1.3 Mounting for right-hand operation



Figure 61

- Mount 90° (or less, depending on the tooth position) rotated to the **left**.

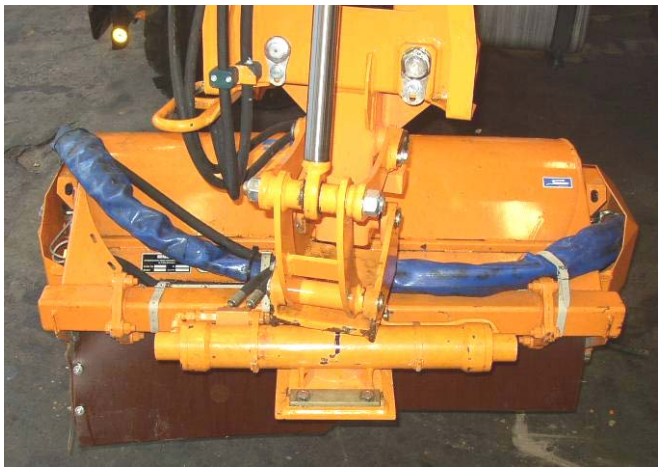
> After rotation, the angle may differ depending on the original position.

- Mount tapered roller bearings, Nilos ring and groove nuts and tighten until the tapered roller bearings are firmly seated.

> The groove nut (Item no.: 165563) is self-locking and must be replaced.

- After conversion, check function and mount the attachment (cutter head) as per the Operating Manual.
- Lubricate the swivel drive.

10.1.1.4 Mounting the attachment from the normal position



- Mounting the attachment from the home position

Figure 62



- After mounting for right-hand operation, the attachment can be swiveled outwards to the limit stop, as shown in the Figure 63

Figure 63



The slewing range has been changed. The hose arrangement must now be checked and adjusted if required.

The original position for left or right-hand operation is changed to the home position by means of conversion in the reverse sequence.

10.2 Tilt compensator

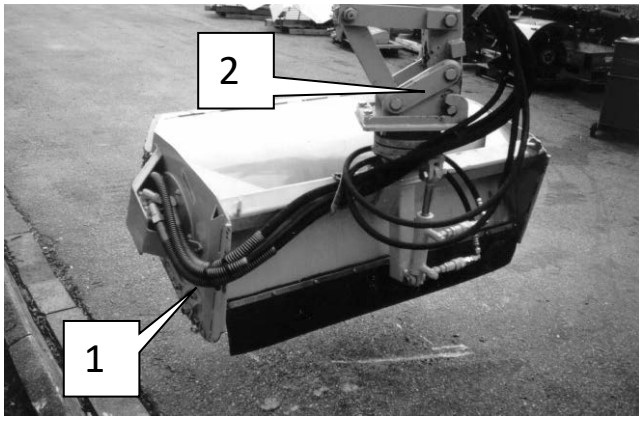


- In each mower head, the ejection angle is altered by using the tilt compensator as the boom is swivelled forwards and simultaneously actuating the **Swivel mower device** function, so that there is no longer adequate protection against ejected matter. If this is not suitably adjusted, there may be a risk of injury (e.g. from stones).
- Always switch off the engine and relieve the hydraulic system before carrying out any work on the embankment mower.



- The tilt compensator is operated via the additional function of the operating panel (see section 3.3).
- Lay the hydraulic hoses in such a way as to prevent them from being damaged during operation.

10.2.1 Adjusting the blade head angle in the forward swivel range



- The tilt compensator is mounted between the blade head holding fixture Figure 64/1 and blade head Figure 64/2.

Figure 64

10.3 Boom offset attachment



- Extreme care must be taken when mounting and dismounting the device in order to avoid serious bruising.
- The boom offset attachment may only be used in conjunction with the obstruction gauge cutter, rough-cutting mower bar or branch and hedging clippers.
- Never use the boom offset attachment without a steel cable for limiting the slewing range of boom 2. Otherwise, boom 2 will impinge onto the opposite lane and it may no longer be possible to lower boom 1.

- Swing boom out to the side and deposit mower device on the ground in the working position so that boom 2 stands vertically on the ground.
- Switch off the boom balancing control system of the mower device.

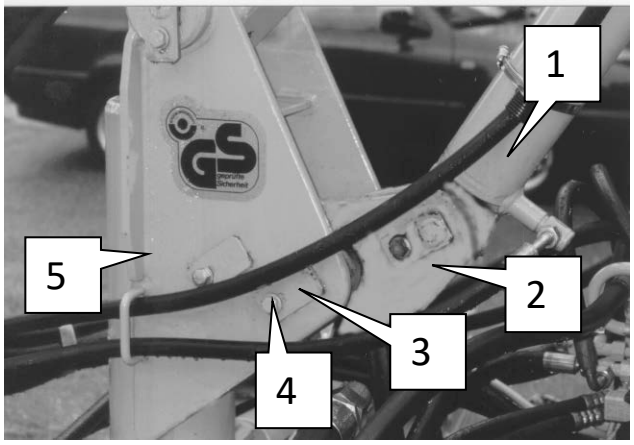


Figure 65

- Reduce the speed of the carrier vehicle to the stationary setting.
- Undo the hex-head screw (Figure 65/4) and pull out the pin (Figure 65/3).
- Fix the boom offset attachment (Figure 65/2) in the centre plate (Figure 65/5) with the pin Figure 65/3.
- Insert the second pin. If this is not possible, dismount the boom offset attachment and align with the inserted adjusting component (eccentric arrangement of bore-hole) so that the bore-holes coincide.
- Secure pins (Figure 65/3) and (Figure 65/6) with hex-head screws (Figure 65/4).
- Carefully adjust the cylinder (Figure 65/1) via the function **”Lower/lift boom”** so that the cylinder locating point Figure 65/7 can be fixed with a pin. Secure the pin with a hex-head screw.

Place the emergency hand lever on the **"Lift"** segment of the valve block. After completing the job, remove the emergency hand lever and keep it safe in the vehicle tool set.

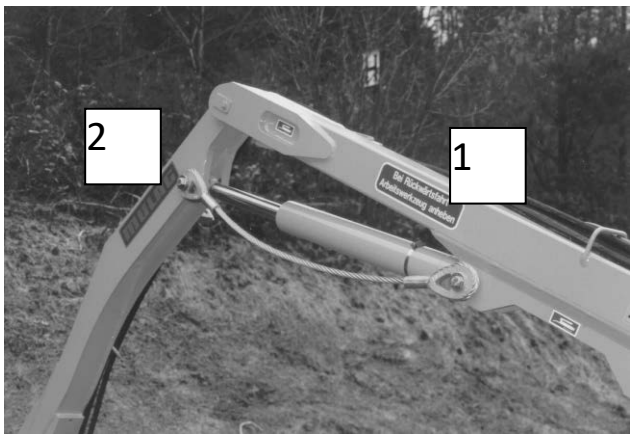


Figure 66

- Mount steel cable (Figure 66/2) for limiting the slewing range of boom 2 parallel to the hydraulic cylinder using two screws and two washers (Figure 66/1).
- To dismount the boom offset attachment, follow the same procedure in reverse order.

10.4 Combined winter operation FME 600 with salt spreader

- In winter the basic equipment can be operated together with a salt spreader when the verge mower has been dismounted or switched-off.

> ☞ The necessity of dismounting the verge mower depends on the loading situation when the spreader has been mounted and filled up (but also when it is empty). The maximal axle loads permitted as well as the minimum load necessary on the rear axle must be observed (see chart chap 10.4.2).

- In working operation, it is possible to change over from the spreader to the mowing equipment, thanks to the hydraulic panel in the driver's cab (see chap.10.4.5). Then, the operator can carry out tree cutting works with a free zone cutter even in winter weather conditions (snow or ice).

10.4.1 Mounting the parking frame

> ☞ The combined operation with mower and spreader makes it necessary to use a universal parking frame, which can be adapted to different spreaders (forms and sizes). However, in some unfavourable cases, some adaptation works may be necessary, even on the universal parking frame.

- With the ball cock situated at the rear of the vehicle, switch off the function of the tilting mechanism.
- Thanks to the installation outside the contour of the tipping stage (on the 4 holding points of the basic frame of the tipping stage / twist locks) the universal parking frame can be used when a spreader has been mounted.



Figure 67

- When mounting the spreader always pay attention that the plates of dispersion remain free.
- Also see following chapter regarding the counter weights in combined used with spreader (chap. 10.4.2).

- Once the spreader has been installed, release – one after the other - the fitting screws on the 4 ball fixtures of the tipping stage and screw the supports (Figure 68/1) under the ball fixtures with the M12 x100 screws.

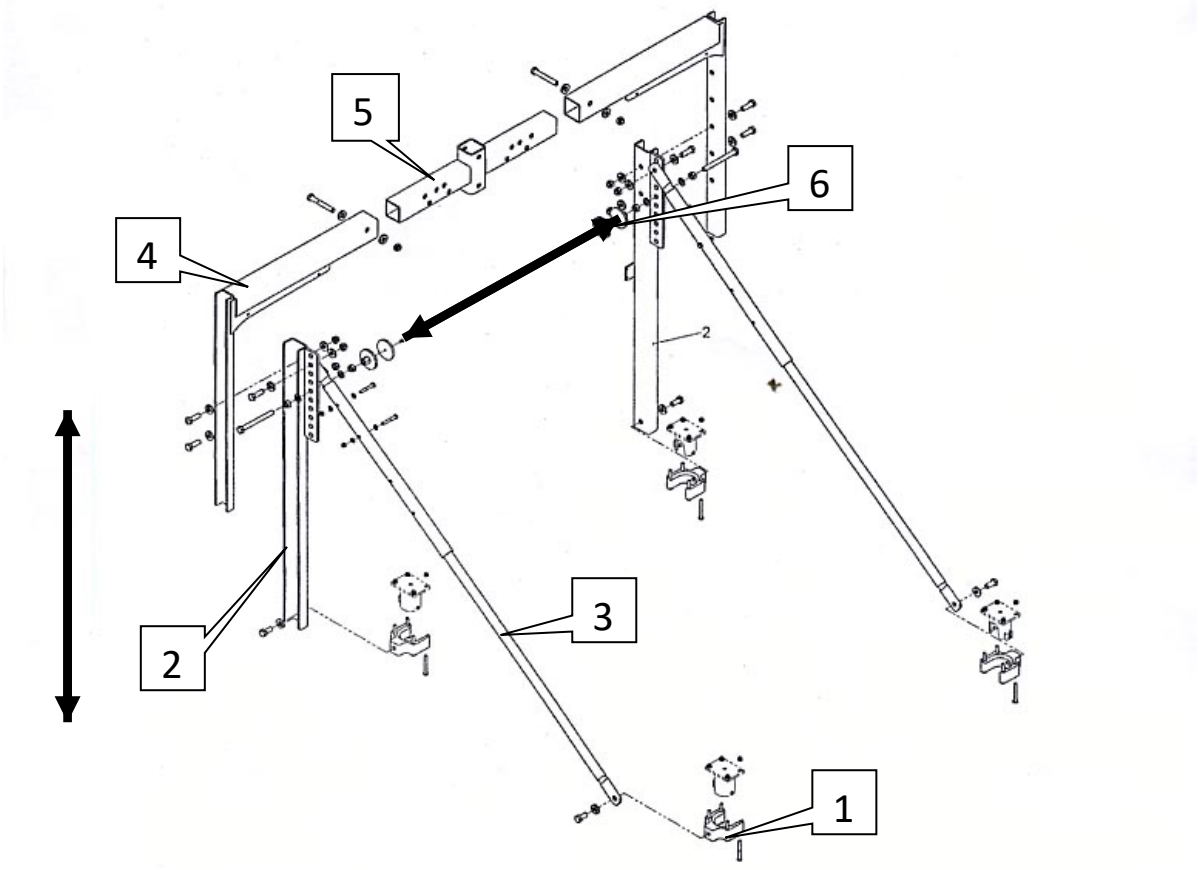


Figure 68

- The universal parking frame is so designed that it can be adjusted in height and width. The height and width adjustments depend on the kind of spreader mounted.
- Mount both holding fixtures (Figure 68/2) and struts (Figure 68/3)

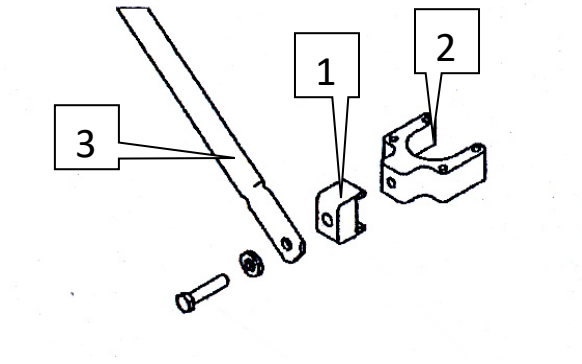


Figure 69

- Depending on the form and construction of the spreader, it may be necessary to mount the 4 setting supports (Figure 69/1) delivered between the holding fixtures (Figure 69/2) and the struts (Figure 69/3).



The 4 setting supports delivered with the universal parking frame (Figure 69/1) are 4,5 cm, so that their installation makes it possible to gain 9 cm for the spreader.

- Screw both holding supports (Figure 68/4) and holding fixtures (Figure 68/5) on both sides.



Figure 70

- Mount both supports (Figure 70/6 and Figure 70) in such a way that they are fixed on a sturdy place of the spreader.

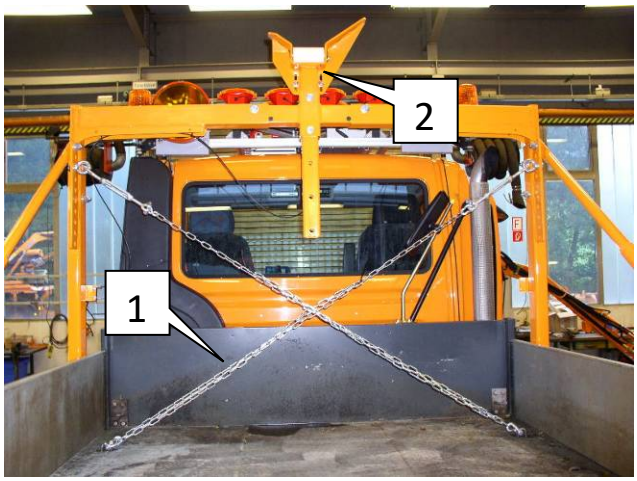


Figure 71

- Should it not be possible, then the anchoring chains (Figure 71/1) delivered with the parking frame must be mounted to give the parking frame the stability necessary.



Figure 72

- When using the anchoring chains, these must be fixed in the securing eyes on the loading surface (Figure 72/1).

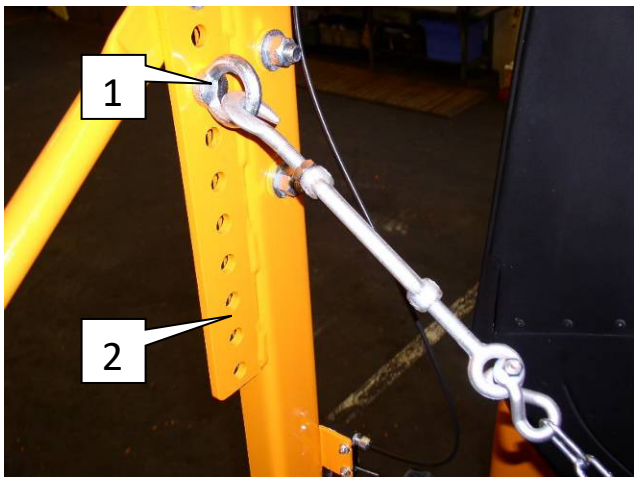
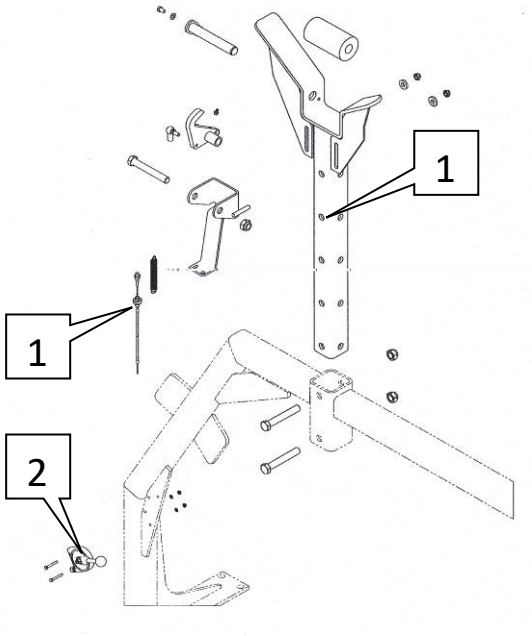


Figure 73

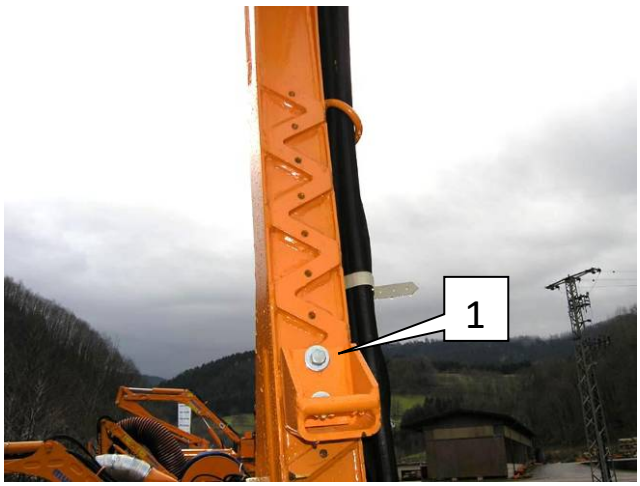
- The Eyebolts (Figure 73/1) delivered with the universal parking frame are used as upper securing points for the anchoring chains. They should be fixed as high as possible on both holding fixtures (Figure 73/2) of the parking frame.

- Install the pre-mounted upper part of the parking frame (Figure 71/2) according to the height of the spreader.



- Mount the tow (Figure 74/1) and the operating lever of the gripping hook Figure 74/2).

Figure 74



- Mount the stop on the boom 2. In transport it must be locked on the parking frame (Figure 75/1).
- The installing place on the boom 2 can be adjusted thanks to the pinholes available.

Figure 75



Figure 76



Figure 77

- Incline the parking frame to the front (Figure 76) or adjust its height so that the loading hatch of the spreader can be opened (Figure 77).



Adjust the height of the parking frame and the stop on the boom 2 so that the vehicle height does not exceed 4m.

Important
The park-in helping mark must be adjusted according to the new heights obtained because of the spreader (see chapter 5.5).

10.4.2 Ballast weight in combined operation with spreader

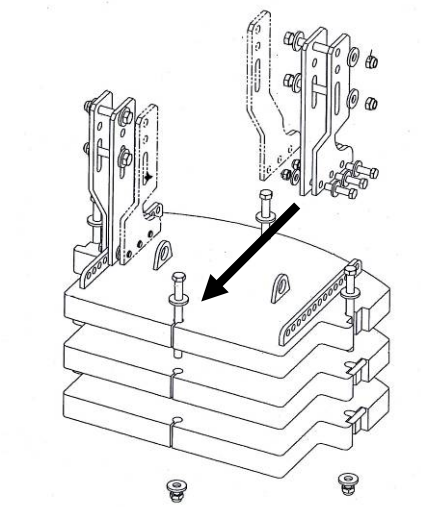


Figure 78

- Thanks to the modular construction, the ballast plates can be removed individually (350 kg each) to obtain the necessary load.



Figure 79

- The weight modules can be displaced in the length to of dispersion remain free.



The ballast weight necessary depends on the carrier vehicle and on the spreader weight. Because of the different possible vehicle weights (depending on the vehicle type, production etc.) and the different types of spreaders, a universal recommendation as regards the ballast weight is not possible. The rear-axle and front-axle loading described in the following charts must be observed and each vehicle must be equipped with the ballast weight according to the real specifications of the vehicle.

10.4.3 Axle loads:

Type of vehicle	Model	Minimum rear axle load		max. rear axle authorised	max. front axle authorised	Code available
		until FIN 1V 201621	from FIN 1V 201621			
U300 K	405.100/101	2300 kg	2300 kg	4600 kg	4600 kg	Serial / X18
U300 K	405.100/101	2300 kg	2300 kg	5500 kg	5100 kg	X19 / X20
U300 L	405.120/121	2300 kg	2300 kg	4600 kg	4600 kg	Serial / X18
U300 L	405.120/121	2300 kg	2300 kg	5500 kg	5100 kg	X19 / X20
U400 K	405.102	2600 kg	2100 kg	7000 kg	6700 kg	X18 / X19
U400 K	405.103	2600 kg	2100 kg	7000 kg	6700 kg	Serial
U400 K	405.103	2600 kg	2100 kg	7000 kg	6800 kg	X18 / X19
U400 L	405.122	2500 kg	2100 kg	7000 kg	6700 kg	X18 / X19
U400 L	405.123	2500 kg	2100 kg	7000 kg	6700 kg	Serial
U400 L	405.123	2500 kg	2100 kg	7000 kg	6800 kg	X18 / X19
U500 K	405.201	2900 kg	2200 kg	8500	7200 kg	Serial
U500 K	405.200/201	2900 kg	2200 kg	9000 **(9500) kg**	7500 kg	X18 **(X19)**
U500 L	405.221	2900 kg	2200 kg	8500	7200 kg	Serial
U500 L	405.220/221	2900 kg	2200 kg	9000 **(9500) kg**	7500 kg	X18 **(X19)**

Origin: Daimler Chrysler (State 01/09)

10.4.4 Hydraulic connections in combined use with spreader

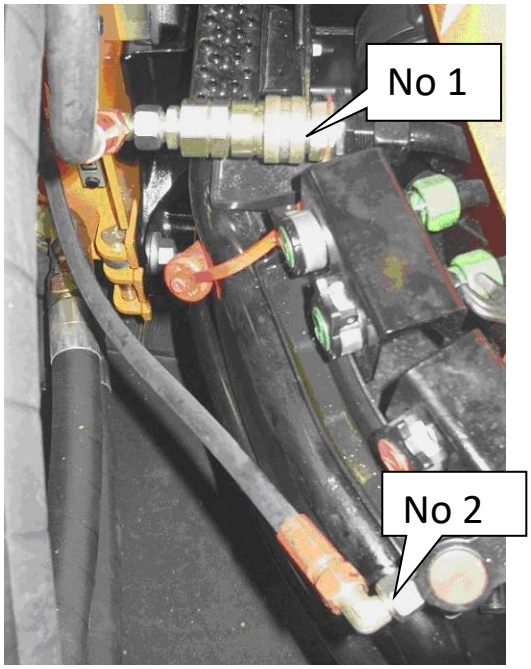


Figure 80 shows the hydraulic connections when the verge mower is mounted.

- No 1 Pressure connection of the basic equipment (FME on circuit 2).
- No 2 Pressure connection of the verge mower on circuit 1, valve 1

Figure 80

- Because of the installation of a salt spreader, the pressure connections of circuit 1 – Valve 1 have to be new connected. This new connection can be carried out thanks to the following description of the connecting adapter, which is used to connect the basic equipment (Figure 81) and of the twin adapter used for the 24 V – socket (Figure 82).

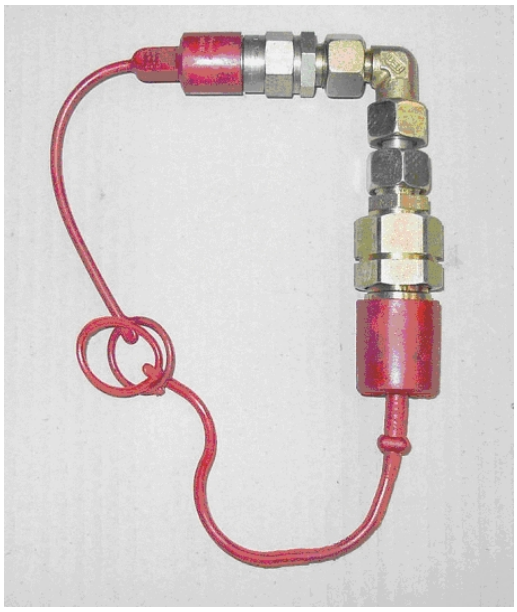


Figure 81

Art- No. 182701



Figure 82

Art.-No. 182702

Der Kupplungsadapter in der flachdichtenden Version (bei Code H 85) hat die Art.-Nr. 200463.



Figure 83

- Once the installation of the salt spreader has been completed, connect the hydraulic hoses to the circuit 2 of the carrier vehicle (according to the operating manual of the spreader) (on the side over the diesel tank)

- Dismount the verge mower or shut it down by removing the pressure connection. Seal the hose pipe (pressure connection of the verge mower) with the protection cap.

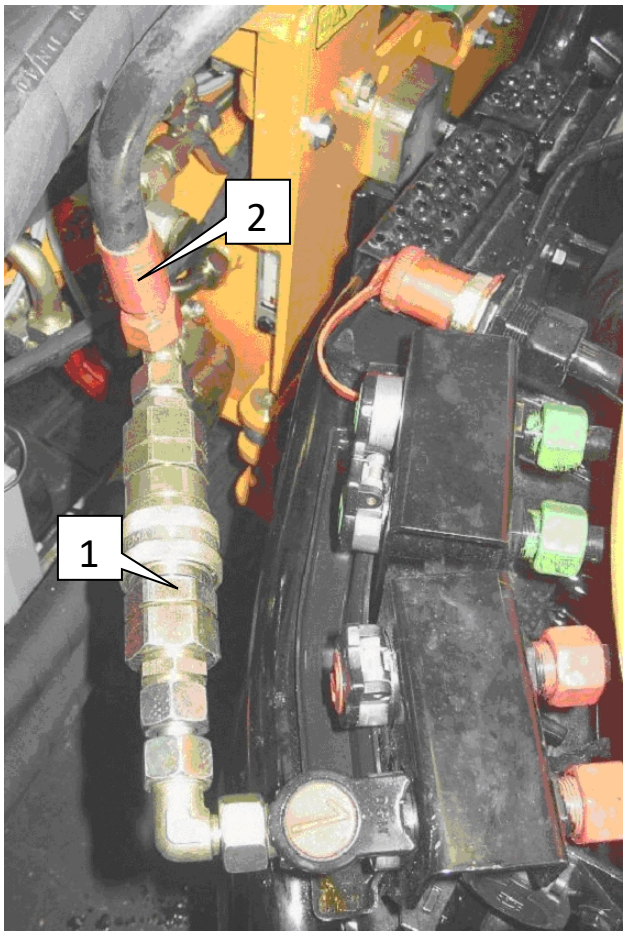


Figure 84

- Plug the coupling adapter (Figure 84) in the hydraulic connection of circuit 1 valve 1.
- Remove the pressure connection of the basic unit (Figure 80/ No. 1) out of the circuit 2 and connect it to the coupling adapter for Circuit 1 - Valve 1 (Figure 84/2).



Figure 85

- Plug the cable for the working current supply of the spreader and basic unit in the twin adapter (Figure 85) and plug connect the adapter to the 24 V – socket of the carrier vehicle (behind the driver's seat).

10.4.5 Switching on / off of the mower and spreader

The mowing equipment and the spreader can be operated according to the following steps.

No .2
Switch on the mowing equipment - Circuit 1, Valve 1 - with switch 1 (Max. 25 l/min)
Attention
Only move the cylinder half so quick and only switch on the mower if necessary.

No. 1
Switch on the hydraulic with switch 1



No. 3
Switch on the salt spreader – Circuit 2 with switch 2 (Max. 50 l/min).

No. 4
X= Change the volu-metric flow between Circuit 1 (25 l/min) and Circuit 2 (50 l/min)

Figure 86

Once the spreading operation has been completed, it is possible to change over to the hydraulic system of the mowing equipment, this via the operating panel of the carrier vehicle.

Because of the modified coupling, the equipments are supplied with hydraulic oil as following:

Circuit II: **Spreader**

Circuit I: **Mowing equipment**

- When operating the mowing equipment with circuit 1, the volumetric flow is slower (max. 25 l/min) and consequently the cylinder movements are slowly
  Thanks to this change over, the boom can be lifted so that the spreader can be filled for winter operation. In this case, the spreader is working with the quicker volumetric flow of circuit II (50 l/min).



- Should the winter operation be stopped or finished and the mower should be used for cutting trees with the free zone cutter for example, it is possible – via the X taste (Figure 86/ No. 4) - to swap the volumetric flow of the spreader (circuit II) and that of the mower (circuit I).
  This changes over makes it possible for the mower to be operated with the maximal working speed when the spreader is switched-off. The mower is then working with the quicker volumetric flow (50 l/min).
- Thanks to this change over, the mower and spreader can be operated without unplugging the hydraulic couplings.

10.4.6 After winter operation

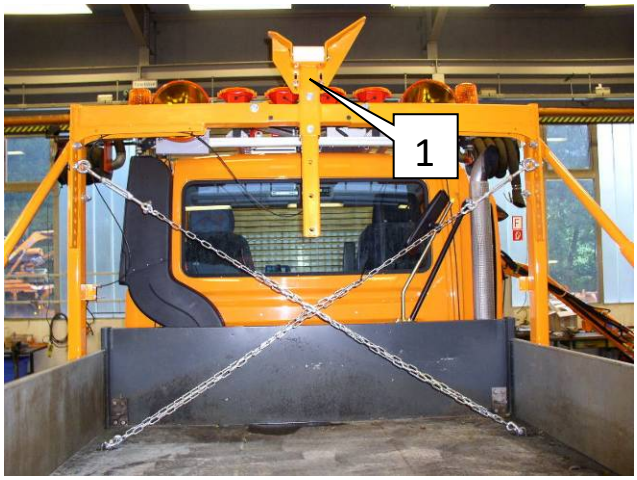


Figure 87

- When the spreader is dismounted after the winter season, the parking frame can remain mounted. The upper part of the parking frame should be put downward in order to reduce transport height. Then other working attachments can be parked too.



Figure 88

- Should no roof structure be mounted, the mowing head can be parked on the platform.

☞	Should the mowing equipment be alone operated - the counter weights – removed for the combined use with the spreader – have to be completed again, in order to obtain the rear and front axle loads necessary. Depending on the operation, the verge mower, which was dismounted, can be re-installed